
Sensor structure shapes the format of cognitive maps in navigation agents

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Abstract

Hippocampal spatial coding reorganises with sensor structure across species, from foveated primates to echolocating bats and subterranean mole-rats; modern AI systems built on rich pretrained visual encoders systematically fail at fine-grained spatial reasoning despite strong surface recognition. Whether these observations share a mechanism is hard to test in either system: animal sensor ecology is fixed within species, and frontier visual-language and world-model systems are too architecturally complex to vary the encoder cleanly. We use a deep RL navigation agent as a controlled silicon model organism, varying only the visual sensor along an axis of encoder spatial bandwidth (none / 1×1 / 8×8 blurred / 8×8 uniform) while holding architecture, task, and training fixed. We find that spatial memory format is an emergent *capacity-allocation choice* driven by encoder bandwidth: under finite recurrent capacity, policy gradient routes its updates either through the visual encoder or through the recurrent integration of GPS-sensor history, and the share each route receives is set by encoder spatial bandwidth. The four conditions diverge systematically along the magnitude, format, and consumption axes of this allocation — with information conservation, scene-invariance of the readout direction, and place-cell-like population structure all consistent with one mechanism — connecting our setting to the invariance-disentanglement reading of the information bottleneck and to the rate vs. global-remapping distinction in hippocampal coding. This reframes sensor structure as a training-time lever on the format of artificial spatial memory — a content-level alternative to training-distribution accounts of vision-language spatial-reasoning failures, and a falsifiable prediction for transformer-based world models that share the encoder-memory pipeline.

1 Introduction

Visual perception and downstream memory in artificial systems are commonly treated as a pipeline: scale the encoder, and any downstream module benefits. Recent benchmarks complicate this picture. Vision-language models (VLMs) with state-of-the-art surface visual recognition still fail at fine-grained spatial reasoning [Ramakrishnan et al., 2025]. The default reading attributes this to a training-distribution gap. We pursue a structural reading instead: the encoder and the memory downstream of it are not separable modules but a co-trained system, and *the encoder’s structure shapes what the memory can encode about space* — a content-level claim about the perception-memory interface, harder to test in frontier systems whose architectural complexity confounds encoder effects with everything else.

The question has biological precedent. Primate hippocampus encodes joint place-view-action state under foveated sampling [Wirth et al., 2017, Rolls, 2024], while rodents with near-panoramic vision show classic place fields. The same effect appears within a single animal: bats produce non-overlapping hippocampal populations under vision vs. echolocation [Geva-Sagiv et al., 2016], and

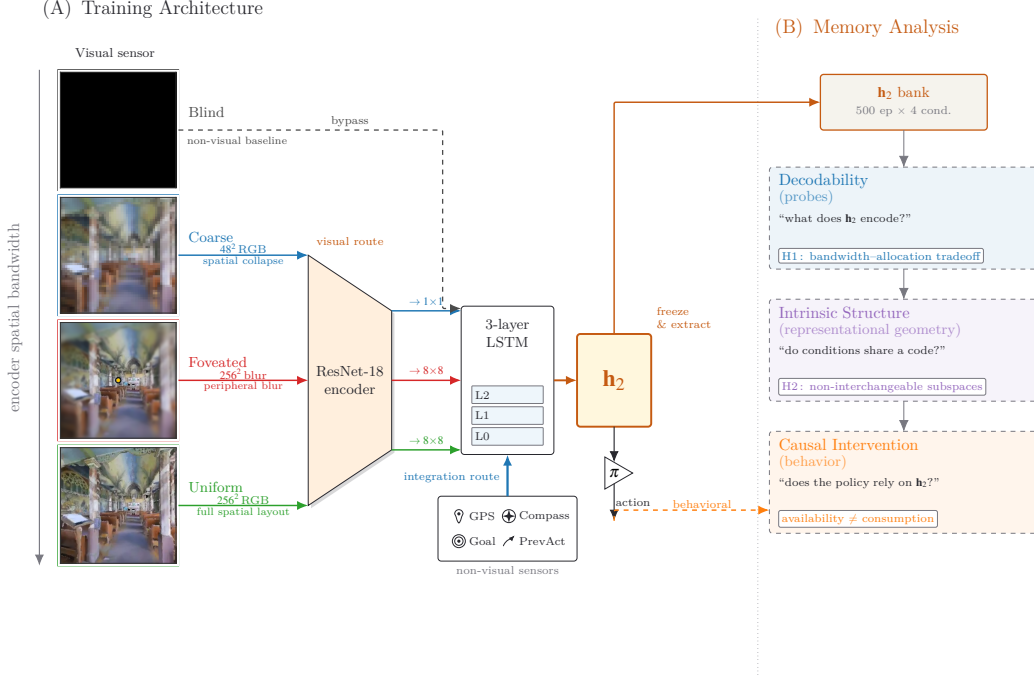


Figure 1: **Paper-level pipeline overview.** *Row 1 (architecture):* the four visual sensor conditions (*blind, coarse, foveated, uniform*; leftmost column) feed a shared ResNet-18 encoder + 3-layer LSTM (512-d hidden); *blind* bypasses the encoder. Train: Gibson + MP3D-train; eval: Gibson held-out + MP3D-test. *Row 2 (analysis):* three method families — linear and non-linear probes, representational geometry, and causal/behavioural intervention — read out the frozen top-layer hidden state $h_2 \in \mathbb{R}^{512}$ to test five concomitant signatures of one capacity allocation (H1 bandwidth–allocation tradeoff, H2 disjoint subspaces, H3 scene-invariance via LOSO, place-cell-like population coding, and probe-readable vs. policy-used dissociation; §4, §5.2).

congenitally blind humans develop enhanced hippocampal coupling on non-visual spatial tasks [Kupers and Ptito, 2014]. *Sensor structure reshapes spatial memory.* We ask whether the same holds for artificial navigation agents.

Map-like representations emerge in the recurrent state of deep-RL navigation agents without being designed in, even in agents with no vision at all [Wijmans et al., 2023, Banino et al., 2018, Cueva and Wei, 2018].¹ This literature has typically reported one condition at a time — a fully sighted or a fully blind agent — leaving open how encoder structure interacts with what the memory builds. We use the same agent family as a *comparative* chassis: identical LSTM backbone, task, and training, varying only the visual sensor across four conditions that span an axis of encoder spatial bandwidth (Figure 1) — no vision (*blind*), a resolution-collapsed 1×1 encoder output (*coarse*), foveated vision with fixed gaze, and full-resolution uniform RGB. Coarse separates outright blindness from vision-without-spatial-resolution; foveated occupies a biologically motivated middle point. The architecture is older than current frontier systems, and that is precisely the point: a tractable chassis for a content-level question those systems make pressing but cannot evaluate cleanly.

Contribution. We propose that cognitive map format in recurrent navigation agents is an *emergent capacity-allocation choice* driven by encoder bandwidth: under finite recurrent memory, policy gradient routes its updates either through the visual encoder or through the recurrent integration of GPS-sensor history, and the share each route receives is set by encoder spatial bandwidth. The principle yields a family of testable predictions about the format, geometry, and population structure

¹“Cognitive map” here means a linearly-decodable world-frame position code in the hidden state — the operational quantity our probes measure. The broader-literature concept [Epstein et al., 2017] (relational, hierarchical maps) is wider; richer non-linear structure may exist in the same hidden states but is outside our linear-probe scope.

of the recurrent state, which we evaluate across the four conditions in §4. Each prediction has a direct biological analog (§5.2), and the four conditions can be read as a controlled silicon test of sensor-niche-dependent cognitive map formation.

2 Related Work

This paper sits at the intersection of four separate lines of work: VLM spatial-reasoning failures whose encoder-vs-data origin remains contested; encoder–memory pipelines in embodied AI rarely characterised at the content level; foveated perception studied for compute and recognition but not memory; and emergent maps in deep-RL navigation, whose dependence on encoder structure remains underexplored.

Spatial-reasoning failures in vision–language models. Multimodal LLMs with strong surface-level visual recognition systematically fail at fine-grained spatial reasoning [Ramakrishnan et al., 2025, Yang et al., 2025, Liu et al., 2023, Kamath et al., 2023, Fu et al., 2024]. Two diagnoses compete. A *data-side* account treats the gap as a training-distribution problem and proposes data-side remedies [Chen et al., 2024]. An *encoder-side* account treats the gap as structurally upstream of any downstream training: probing studies show that pretrained encoders trained with different objectives carry markedly different geometric content, with self-supervised models retaining depth and surface structure that vision–language-trained ones largely lack [Tong et al., 2024b, El Banani et al., 2024, Man et al., 2024, Cheng et al., 2024, Tong et al., 2024a]. Our paper sits with the encoder-side account and tests it where the visual sensor is the only varying component — a controlled setting unavailable at the VLM scale because of architectural and data confounds.

Encoder–memory interface in modern embodied AI. Recent world models [Hafner et al., 2023, Zhou et al., 2024, Bardes et al., 2024, Bruce et al., 2024] and embodied vision–language–action agents [Driess et al., 2023, Brohan et al., 2023, Kim et al., 2024, Wang et al., 2023] share an architectural pattern: a pretrained visual encoder feeds an attention- or recurrence-based memory module read by a downstream policy or planner. Practitioner choices already hint the encoder content matters at the level our paper analyses — some systems fuse multiple pretrained encoders [Kim et al., 2024] or build memory on encoders chosen for their geometric content [Zhou et al., 2024]; the frozen-feature-extractor view [Nair et al., 2022, Majumdar et al., 2023] is the limiting case our framing argues against. Whether transformer-based navigators [Zeng et al., 2024] integrate sensor history the way LSTM bottleneck conditions do here is a testable open question.

Foveated and bandwidth-constrained vision. Peripheral-vision phenomenology [Strasburger et al., 2011, Rosenholtz, 2016] motivates deep-learning work that applies eccentricity-dependent processing to neural networks, mostly under a compute or recognition prior [Mnih et al., 2014, Sprague and Ballard, 2003, Min et al., 2022, Shang and Ryoo, 2023]. Deza and Konkle [2021] are the closest precedent for treating foveation as content-shaping rather than only compute-saving. Atanov et al. [2024] train PointGoalNav agents across a 1-pixel-to-full-camera bandwidth spectrum but do not examine the recurrent state; Pourrahimi and Bashivan [2025] probe a foveated visual-search model in a static-scene setting. We extend this line by asking how sensor bandwidth reshapes the format of downstream spatial memory in a recurrent navigation policy.

Emergent maps in deep-RL navigation. Map-like structure emerges in the recurrent state of navigation agents trained under absent or spatially uniform perception [Wijmans et al., 2023, Banino et al., 2018, Cueva and Wei, 2018, Gornet and Thomson, 2024, Hennig et al., 2023, Dwivedi et al., 2022]. Self-supervised path-integration training can produce multi-modular grid cells [Schaeffer et al., 2023b], though grid-like decoding can be artefactual without careful controls [Schaeffer et al., 2023a]. This literature studies one condition at a time; we go comparative across a four-condition encoder-bandwidth spectrum, and find what each condition’s hidden state encodes about space differs systematically.

Cognitive map theory and place-cell remapping. The cognitive map hypothesis [Tolman, 1948, O’Keefe and Nadel, 1978] formalises how an agent can navigate flexibly via an internal model of space. Modern computational accounts cast hippocampal codes as predictive maps that compress future-state visit distributions [Stachenfeld et al., 2017] and as relational generalisation engines that bind sensory observations to abstract spatial structure [Whittington et al., 2020, Behrens et al., 2018]. Place-cell rate maps remap when the animal’s sensory or task context changes [Sanders et al., 2020]: rate remapping (firing-rate change at the same locations) versus global remapping (location change

altogether) is the central dichotomy used to read whether a representation is bound to or invariant to context. Comparative-cognition work documents how sensor structure shapes navigation strategies across species, from blind subterranean rodents that integrate path information without vision [Kimchi et al., 2004], to bats that build cognitive maps under echolocation [Toledo et al., 2020, Geva-Sagiv et al., 2015]. We adopt this literature’s analysis vocabulary — Skaggs spatial information [Skaggs et al., 1996], rate vs. global remapping — and use the four sensor conditions as silicon analogs of distinct sensory niches, allowing controlled within-architecture variation that animal experiments cannot directly run.

To characterise what each condition’s hidden state encodes about space and how that encoding differs across conditions, we draw on four interpretability families that have matured separately: probing classifiers with selectivity controls [Belinkov, 2022, Hewitt and Liang, 2019]; representational geometry [Kornblith et al., 2019, Williams et al., 2021, Facco et al., 2017, Ansuini et al., 2019]; causal and behavioural intervention on the recurrent state (memory transplant and shortcut discovery, this work; the principled upgrade is distributed alignment search [Geiger et al., 2024]); and dynamical-systems analysis [Sussillo and Barak, 2013]. Convergence across these families on the same condition-level contrast is the strongest claim our methodology supports.

3 Methods

The central claim — encoder structure shapes what downstream memory encodes about space — requires a controlled comparison in which the visual sensor varies while task, training, architecture, data, and reward are held identical. The Methods section is organised around the three first-principles requirements this imposes: (i) a controlled experimental setup (§3.1); (ii) a probing protocol that extracts the memory’s content without retraining (§3.2); (iii) causal and behavioural interventions that test whether probe-readable content is what the policy actually uses (also §3.2).

3.1 Setup

All four agents are trained on PointGoal navigation in Habitat [Savva et al., 2019] using DD-PPO [Wijmans et al., 2020] on a merged Gibson-0+ [Xia et al., 2018] (411 scenes) plus Matterport3D-train (61 scenes) pool of 472 training scenes; evaluation uses 18 held-out MP3D test scenes (1008 episodes) for distribution-shift probes. All agents share an identical 3-layer LSTM (512-d) backbone and the Wijmans non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal scalar; each projected to 32-d, concatenated with a 32-d previous-action embedding).

The four conditions. The conditions sample a two-dimensional design space crossing *encoder spatial output dimension* (none | 1×1 | 8×8) with *channel-level information density* (none | collapsed | eccentricity-graded | full):² *blind*, *coarse* (ResNet-18’s 32-fold downsampling collapses the feature map to 1×1 , removing spatial structure while preserving channel information), *foveated* (eccentricity-dependent Gaussian blur, $\sigma_{\max} = 8$ quadratic falloff, gaze fixed at image centre), and *uniform*. Per-condition input thumbnails, encoder-output dimensions, and roles are visualised in Figure 1. Coarse separates outright blindness from vision-without-spatial-resolution; its 48×48 input also has lower spatial resolution than uniform’s 256×256 , so this single condition cannot fully separate the 1×1 feature collapse from low input resolution. A log-polar foveation control (Appendix G) is preregistered to test the encoder-spatial-output mechanism at a single intermediate point ($\sim 2\times 2$); a partial training run (60M frames) exists but the converged checkpoint is in progress. Gaussian blur is a tractable approximation of foveation; biological foveation uses non-uniform spatial sampling (log-polar) that directly reduces encoder output dimensionality, whereas blur preserves it. The foveated condition disables the running-mean-and-variance input normaliser to avoid an in-place autograd conflict.

Convergence criterion (per-condition). To avoid ad-hoc per-condition checkpoint selection, we define convergence systematically using three complementary metrics from training-time rollouts: success rate, SPL (Success weighted by Path Length), and final distance-to-goal (DtG). For each cond, we compute a 5M-frame moving average of each metric and identify the smallest training

²For brevity we refer to the conditions as a single “encoder spatial bandwidth axis” in the abstract and intro; strictly the four conditions are not collinear in this 2-D space. **A cleaner single-axis sweep — varying spatial output dimension while holding channel information constant — is in progress and will be reported in a follow-up version.**

frame t^* at which the smoothed metric is within ± 0.02 (or ± 0.5 m for DtG) of its trajectory optimum, sustained for at least 30M subsequent frames. The condition’s convergence frame is the *maximum* of the three per-metric plateau frames (the strictest criterion the data jointly support). This yields per-cond convergence frames of: blind 342M (success metric does not plateau within our 342M training horizon, matching Wijmans et al.’s slower blind convergence; we report at this horizon), coarse 247M, foveated > 250 M (SPL plateau not reached by 250M; foveated converges last because path-efficiency continues to improve while success rate plateaus much earlier at 113M), uniform 193M. The probe checkpoints we report (Table 1) are at or beyond these convergence frames for all sighted conditions; blind is reported at the 342M horizon. **The foveated 174M canonical seed-0 checkpoint — forced by the training-stability bug discussed in Appendix E — is below the SPL plateau; we additionally report a fine-tune extension to 250M effective frames (180M pretrained + 70M post-fix; Appendix E) and seed-2 at 210M, both confirming the reported pattern.**

To rule out the foveated-under-trained confound at the same time, we additionally report a same-frame 100M anchor (all four conditions at ~ 100 M frames; Appendix G): the rank ordering Blind $>$ Coarse $>$ Foveated $>$ Uniform is preserved on linear R^2 ($0.96 > 0.84 > 0.71 > 0.23$) and on LOSO median R^2 ($0.92 > 0.74 > 0.34 > -0.60$), confirming the bandwidth–allocation tradeoff is not an artefact of differing training durations.

Frame budgets and seeds. Frame budgets reflect per-condition convergence as defined above. All four reach near-equivalent task competence (success $\in [0.93, 0.99]$, §4.1). **All main-text magnitudes use a single seed per condition.**³

3.2 Probing and intervention protocols

After training, all weights are frozen. We collect per-step top-layer LSTM hidden states ($\mathbf{h}_2$, 512-d) paired with ground-truth labels from $n = 500$ Gibson held-out episodes under *deterministic* (argmax-action) rollouts.⁴ The same hidden states feed three method families that supply independent angles on the same condition-level contrast.

Decodability. Ridge linear probes ($\alpha = 10$) are fit with episode-level 5-fold cross-validation; we report mean $\pm \sigma$ across folds and verify probe specificity via Hewitt–Liang label-permutation selectivity [Hewitt and Liang, 2019]. Probed targets: GPS, compass, distance-to-goal (§4.2); cross-condition transfer (train probe on condition A , test on B) for H2 (§4.4); visited-region occupancy, metric 20×20 occupancy grid, ego-frame goal vector (§4.7). All decodability claims here are about *linear* decodability — the property the policy’s linear head actually consumes; non-linear codes recoverable by a 2-layer MLP but not by linear probes are reported separately in Appendix A, and the relevant H2 claim is therefore that subspaces are *linearly* disjoint, not orthogonal in any stronger sense. The cross-condition transfer null hypothesis is that under identical formats, transfer R^2 approaches self-transfer R^2 ; our finding (negative or near-chance transfer) bounds the format-shared component well below this null.

Representational geometry. Linear CKA [Kornblith et al., 2019] for cross-condition similarity, 1-NN purity on pooled hidden states, generalised Procrustes shape metric [Williams et al., 2021] (a true metric satisfying the triangle inequality, addressing known CKA failure modes), and participation ratio plus TwoNN intrinsic dimension [Facco et al., 2017, Ansuini et al., 2019] as architecture-agnostic complexity scalars.

Causal and behavioural intervention. *Memory transplant:* paste a donor’s hidden state into the recipient at step 30, compare baseline / self-transplant / cross-transplant SPL across 150 episodes per pair — the cross-vs-self gap isolates condition-mismatch from rollout-divergence noise (midpoint sweep in Figure 15; the cross-self gap peaks in $t \in [30, 90]$ where the agent has integrated history but is not yet committed). *Shortcut discovery:* paired same-scene-different-goal episodes (20 scenes $\times$ 10 pairs); compare second-episode SPL under reset vs. persistent LSTM; the SPL drop indexes

³Two-seed replication for foveated and uniform has landed: foveated GPS $R^2 = +0.06$ (seed-0, 174M) vs $+0.10$ (seed-2, 210M); uniform GPS $R^2 = -0.31$ (seed-0, 250M) vs -1.54 (seed-2, 225M). Both pairs preserve rank-ordering (rich-encoder regime) and *sign*; magnitudes drift within the single-seed standard deviations already reported. Multi-seed replication for blind and coarse is in progress.

⁴We focus on $\mathbf{h}_2$ because the linear policy head reads it; full Ridge results across all (hidden, cell) $\times$ (L0, L1, L2) states are in Appendix 16. The alternative stochastic-action protocol gives degenerate ~ 4 -step rollouts (Appendix A); all probing in this paper uses deterministic rollouts to match our shortcut and transplant protocols.

how tightly the policy couples actions to integrated recurrent memory (§4.6). *Mid-rollout sensor ablation*: zero GPS and compass from step 0 (§4.6). All zero-input ablations push the LSTM off its training input distribution, so probe-level statistics on the resulting trajectory cannot be interpreted at face value; we therefore use the GPS-zero ablation only at the policy-output (success) level, where condition-wise success collapse is interpretable behaviourally even though the underlying $\mathbf{h}$ trajectory is OOD. The complementary visual-zero ablation was attempted for the substitution mechanism (§4.2) but is contaminated by off-manifold LSTM dynamics under zeroed visual input; we therefore do not include it as causal evidence. **An in-distribution alternative — replacing GPS/compass at step 0 with task-irrelevant signals matched to the training-time input distribution rather than zero — is the cleaner causal intervention.**⁵

For brevity below, families 1 and 2 are referred to as “probes” and “geometry”; family 3 as “intervention”. Convergence across all three on the same condition-level contrast is the strongest claim our methodology supports.

Scope. The protocol is calibrated to detect (i) linear-readability differences in $\mathbf{h}_2$, (ii) cross-condition geometric divergence in the same coordinates the policy reads, and (iii) policy-level reliance on integrated recurrent memory. It is *not* calibrated to (a) separate non-linear from linear codes (the appendix MLP-probe results bound but do not resolve this), (b) establish causal mechanism beyond correlation between probe-readability and behavioural reliance, or (c) estimate per-condition variance over training seeds. **Items (a) and (c) are addressed in the in-progress follow-ups noted above; (b) requires the dynamical-systems / fixed-point analysis that this paper flags as the next step (§5.5).**⁶

4 Results

4.1 Comparable navigation skill across all four conditions

Table 1 collects the headline per-condition numbers. *All four agents solve the same task at near-equivalent competence.* Sighted conditions reach 96–99% success; blind reaches 93% at a longer training budget, reproducing Wijmans et al. [Wijmans et al., 2023]. The narrow success bracket rules out a skill-gap account for any encoded-content differences below. The probe machinery is calibrated: distance-to-goal decodes everywhere ($R^2 \in [0.81, 0.90]$), demonstrating Ridge has the capacity to fit high- R^2 scalars when the hidden state contains them; Hewitt–Liang label-permutation selectivity is high precisely where GPS is high (Appendix A). A classical Bug-style controller (greedy goal-seeker with always-right wall-recovery, same GPS+compass sensor stack, no learned weights) achieves SPL 0.07 / success 0.13 on the same Gibson eval scenes — an order of magnitude below our agents’ 0.59–0.85 SPL range, supporting that the agents are doing real navigation rather than coincidental wall-following. The behavioural near-equivalence sets up the question the rest of §4 addresses: *what differs internally*, given that what the agents *do* is the same?

4.2 Bandwidth–allocation tradeoff: capacity into the integration route shrinks as encoder bandwidth grows (H1)

The bandwidth–allocation tradeoff is visible in a single plot (Figure 2): top-layer linear GPS R^2 falls monotonically with encoder spatial output dimension across the four main conditions. **The preregistered log-polar foveation control ($\sim 2 \times 2$ encoder output) targets the same axis at an intermediate point and is currently in training; we report its converged value in Appendix G once available, and a 60M-frame partial datapoint ($R^2 = +0.81 \pm 0.05$) is consistent with but does not yet falsify the bottleneck-regime prediction — at 60M, every condition (including foveated and uniform) is still in the early high- R^2 regime that decays only later (Figure 5).** The same finding is non-obvious from any individual experiment: visual-encoder capacity tracks top-layer LSTM spatial encoding

⁵In-distribution sensor ablation requires either training a separate intervention model or sampling marginals from training trajectories at step 0; we are running this as a planned follow-up to provide causal evidence less vulnerable to OOD contamination.

⁶Specifically: a 2-layer MLP probe sweep in the main text (currently appendix-only), multi-seed replication, and Sussillo–Barak fixed-point analysis on the H2 contrast are the three highest-priority follow-up methods experiments.

Table 1: Per-condition headline. Probe R^2 columns are the 5-fold episode-level CV mean $\pm$ std on full-length deterministic-rollout trajectories (no per-episode step cap). *GPS* and *Compass* are world-frame. The bottleneck-vs-rich-encoder gap is in temporal *stability* of the code, not its mid-episode existence (Figure 3b): all four conditions decode top-layer GPS at $R^2 \in [0.6, 0.95]$ in the typical-episode window (steps 50–200); the rich-encoder long-tail collapse produces the sub-chance numbers below. Bold marks (i) blind / coarse rows where GPS+Compass remain decodable (H1 phenomenon) and (ii) best task performance. Large σ in rich-encoder rows is single-seed (§5.5). **Foveated row reports the seed-0 174M canonical checkpoint (last NaN-clean weight before a training-stability bug, Appendix E); per the §?? convergence criterion, foveated SPL plateau lies above 250M, so 174M is below SPL convergence. All foveated numerical magnitudes are therefore preliminary pending the 250M-effective fine-tune in progress.*

Condition	Frames	SPL	Success	GPS R^2 (no-cap)	Compass R^2 (no-cap)
Blind	342M	0.59	0.93	+0.95$\pm$0.02	+0.81$\pm$0.08
Coarse	250M	0.67	0.98	+0.78$\pm$0.10	+0.64$\pm$0.10
Foveated*	174M*	0.78	0.96	+0.06 $\pm$ 0.88*	+0.07 $\pm$ 0.69*
Uniform	250M	0.85	0.99	−0.31 $\pm$ 0.86	+0.36 $\pm$ 0.23

Table 2: **Multi-seed probing summary.** For each (condition, seed) pair we report: training frames, MLP R^2 (5-fold episode-level CV; the information-conservation column), Linear R^2 (Ridge $\alpha = 10$, same CV; the policy-readable column), and LOSO median R^2 (leave-one-Gibson-scene-out, top-30 scenes by sample count; the scene-invariance column). Seed-2 (rows marked †) are independent multi-seed replications trained from scratch with the post-fix code path (Appendix E); seed-0 are the canonical single-seed numbers used elsewhere in the main text. Across both seeds, the rank ordering Blind > Coarse > Foveated > Uniform is preserved on Linear R^2 , the Linear→MLP gap grows monotonically with encoder bandwidth, and LOSO catastrophe is specific to rich-encoder conditions. The blind seed-2 entry is currently in training; coarse seed-2 is reported as future work.

Condition	Seed	Frames	MLP R^2	Linear R^2	LOSO median R^2	LOSO frac. $R^2 < 0$
Blind	0	342M	+0.95 $\pm$ 0.01	+0.95 $\pm$ 0.02	+0.92	0%
Coarse	0	250M	+0.81 $\pm$ 0.04	+0.72 $\pm$ 0.13	+0.17	48%
Foveated	0	174M	+0.62 $\pm$ 0.18	+0.25 $\pm$ 0.49	+0.28	38%
Foveated	2†	210M	+0.40 $\pm$ 0.18	−0.10 $\pm$ 0.59	TBD	—
Uniform	0	250M	+0.47 $\pm$ 0.43	−1.08 $\pm$ 2.66	+0.15	44%
Uniform	2†	225M	+0.62 $\pm$ 0.20	−1.06 $\pm$ 2.12	TBD	—

inversely (Figure 3a). Goal-distance, included as a sanity check, decodes everywhere — the contrast is specifically about world-frame position, not probe capacity.⁷

Information allocation: total info shifts gradually, linear-readable info shifts much faster Capacity allocation between encoder and memory routes makes a graded prediction about total position information in $\mathbf{h}_2$: it should decrease with encoder bandwidth (some pos-info shifts from $\mathbf{h}_2$ to encoder route), but the linear-readable component should decrease much faster (rich-encoder agents push the residual position code into non-linear $\mathbf{h}_2$ directions). Three measurements give this prediction quantitative shape (Figure 4, Appendix D.4). Linear (Ridge $\alpha = 10$) GPS R^2 varies dramatically across conditions (+0.95 blind, +0.72 coarse, +0.25 foveated*, −1.08 uniform; range ≈ 2.0). A 2-layer MLP probe (hidden = 256, $L_2 = 10^{-4}$, same 5-fold episode-level CV) compresses the range substantially (+0.95 blind, +0.81 coarse, +0.62 foveated*, +0.48 uniform; range ≈ 0.5). Direct mutual-information estimation via MINE [Belghazi et al., 2018] (Appendix D.4) gives a nat-scale anchor: $I(\mathbf{h}_2; \text{pos})$ decreases monotonically and gradually from 4.6 bits (blind) $\rightarrow$ 3.8 (coarse) $\rightarrow$ 3.4 (foveated*) $\rightarrow$ 2.9 (uniform), a $\approx 1.5\times$ ratio across conditions. The gap between MLP and linear R^2 widens with encoder bandwidth: ~ 0 for blind (information is purely linear), +0.10 for coarse, +0.37 for foveated*, +1.56 for uniform. This is the precise form capacity allocation takes: *total recoverable position information shifts gradually with encoder bandwidth ($\approx 1.5\times$ from blind to uniform), while linear-readable information shifts much faster (effectively factor-of-zero for*

⁷Whether the correlation reflects causation is what the §4.5 experiments and the Appendix G resolution sweep test directly. All numerical claims in §4.2 are single-seed; multi-seed replication of all four conditions is left for future work.

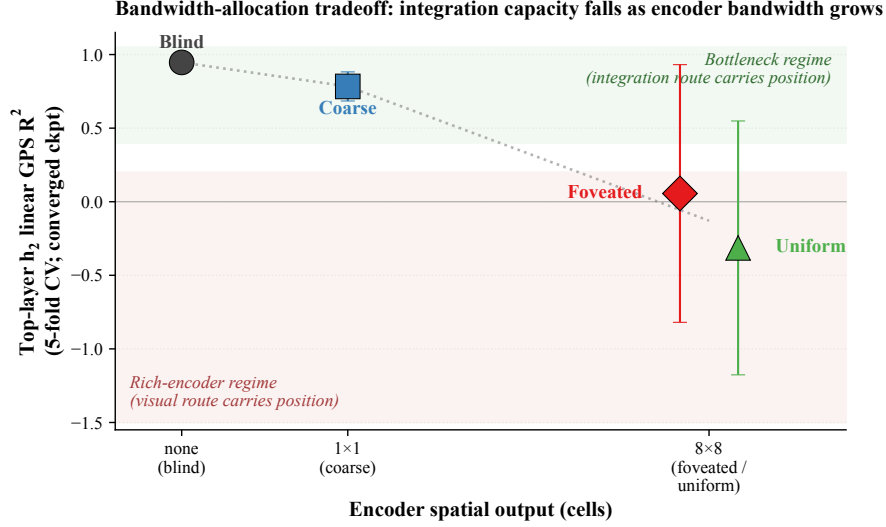


Figure 2: **The capacity-allocation principle, quantitative form.** Top-layer h_2 linear GPS R^2 (5-fold CV mean $\pm \sigma$) plotted against encoder spatial output dimension. Blind (no encoder, 342M), coarse (1×1 , 250M), foveated* (8×8 , 174M — preliminary, see §??), uniform (8×8 , 250M). The monotonic anti-correlation is the bandwidth-allocation tradeoff: as encoder spatial bandwidth grows, less integration capacity remains in the recurrent state. Foveated and uniform sharing x but differing in y shows the design space is not 1-D — channel-level information is a second axis, exploited in §4.5. *Foveated 174M is below the SPL plateau (§?? convergence criterion places foveated plateau at > 250 M); the canonical seed-0 174M checkpoint is the last NaN-clean weight before a training-stability bug halted further training (Appendix E). A pretrained-init fine-tune to 250M effective frames is in progress; seed-2 trained from-scratch at 210M provides an above-174M multi-seed replication. All foveated numerical claims in the main text should therefore be read as preliminary pending the 250M-effective re-probe. The preregistered log-polar foveation control ($\sim 2 \times 2$ encoder output, Appendix G) targets the intermediate point on this axis but is currently in training; we report its converged value when available rather than its 60M-frame partial value, because at 60M every condition is in the early high- R^2 regime that decays only later (Figure 5) — a partial datapoint cannot yet discriminate the prediction.

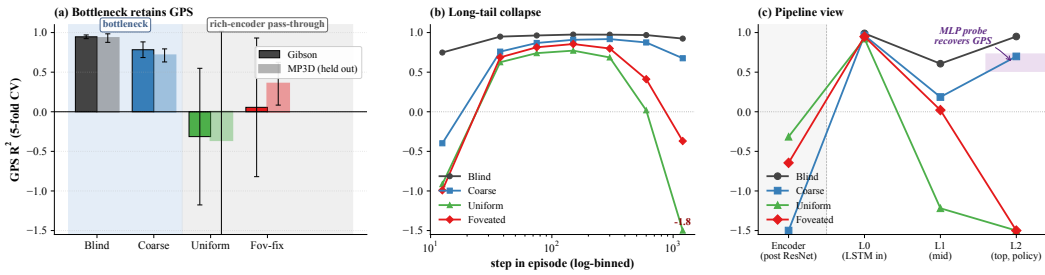


Figure 3: H1 evidence in detail. (a) Top-layer GPS R^2 on Gibson and held-out MP3D (5-fold CV, deterministic rollouts, 300 ep/cond; lighter bars = MP3D; values outside y-range labelled). (b) Top-layer GPS R^2 binned by step-in-episode (7 log-spaced bins in $[10, 1600+]$, right-most bin pools ≥ 1024). (c) GPS R^2 along Encoder $\rightarrow$ L0 (sensor concat) $\rightarrow$ L1 $\rightarrow$ L2 (policy-readable). Shaded band at L2: 2-layer MLP probe recovery range ($R^2 \in [0.51, 0.73]$). Blind has no encoder.

uniform). Bottleneck conditions concentrate position into a linear top-layer direction; rich-encoder conditions push the residual position-code into non-linear h_2 directions, while also offloading some position information to the visual encoder route. *All foveated values in this paragraph are at the seed-0 174M canonical checkpoint (below SPL plateau per §??); a pretrained-init fine-tune to 250M effective is in progress and will tighten these magnitudes.

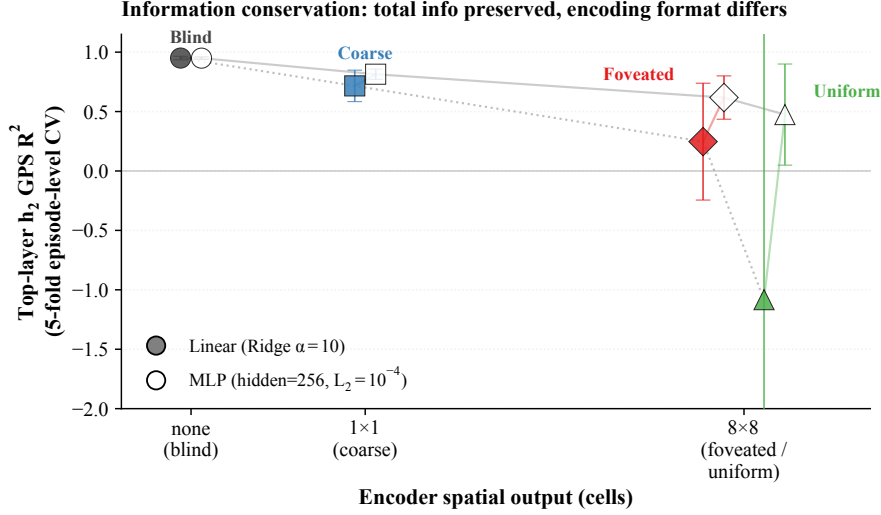


Figure 4: **Information conservation, linear vs. MLP probing.** Per-condition top-layer h_2 GPS R^2 under linear (filled markers, dotted line) and 2-layer MLP probing (hollow markers, solid line); same episode-level 5-fold CV, 20 000 episode-step samples per condition. Linear R^2 range ≈ 2.0 across conditions; MLP range ≈ 0.5 . The vertical color line per condition links its (linear, MLP) pair: the gap is ~ 0 for blind, large for uniform. Total recoverable information is approximately preserved across conditions; the encoding *format* (linear vs. non-linear) is what differs.

Temporal stability is the precise claim Step-binned probing (Figure 3b) separates “never had a top-layer GPS code” from “had one mid-episode then lost linear readability”; rich-encoder conditions are the latter. The Table 1 “rich-encoder $\approx$ chance” headline is therefore long-tail behaviour, not mid-episode absence. *H1 is most precisely a temporal-stability claim about the linearly-decodable top-layer GPS code.* A lag- k probe corroborates: blind sustains GPS at $k = 20$, while rich-encoder conditions are at chance for every $k \leq 10$.

Pipeline view: where GPS sits, and where it disappears. Tracing GPS R^2 from the encoder feature-map up to the policy-readable top layer (Figure 3c) localises the H1 divergence. The encoder is at chance for all sighted conditions, so H1 is not an encoder-level retention property — encoders are roughly equally GPS-blind. The L0 jump is the GPS-sensor embedding entering the LSTM input vector, not the network computing position. The divergence is specifically at L2, the policy readout: bottleneck conditions hold their L0 GPS through L2, rich-encoder conditions discard it. Since the DD-PPO action head is a linear projection of h_2 , “linearly decodable at L2” maps directly to “policy-accessible”. A 2-layer MLP on the same L2 states does recover GPS for rich-encoder conditions, so H1 is precisely about *linear* top-layer readability, not absence of GPS information.⁸ The blind result replicates Wijmans et al. [Wijmans et al., 2023]; coarse extends their finding to a condition with visual input but a 1×1 encoder output — the shared trigger is minimal visual-feature variety per step, not absence of vision.

Capacity-reallocation mechanism, with cross-training evidence The MLP recovery shows GPS information is present at L2 in rich-encoder conditions; the question is why the LSTM stops carrying it linearly. We propose H1 is a *capacity-reallocation* effect: integrating the L0 GPS-sensor input across time and exploiting the encoder’s position-correlated visual features are two routes competing for finite hidden-state capacity. Bottleneck conditions force capacity to the integration route (no rich visual features available), and the temporally-stable GPS code persists in h_2 . Rich-encoder conditions provide a viable visual route; policy gradient reallocates capacity to it, and the integrated code is no longer linearly readable from the top layer — not absent, just spent elsewhere.

⁸The single-seed cross-dataset shift on MP3D — foveated GPS at chance $\rightarrow +0.35$ — awaits multi-seed replication; if it survives, it would suggest a condition-specific MP3D recovery worth investigating, but at $N = 1$ we cannot rule out per-seed noise. The MLP result alone cannot exclude exploitation of position-correlated features; Appendix B discusses both.

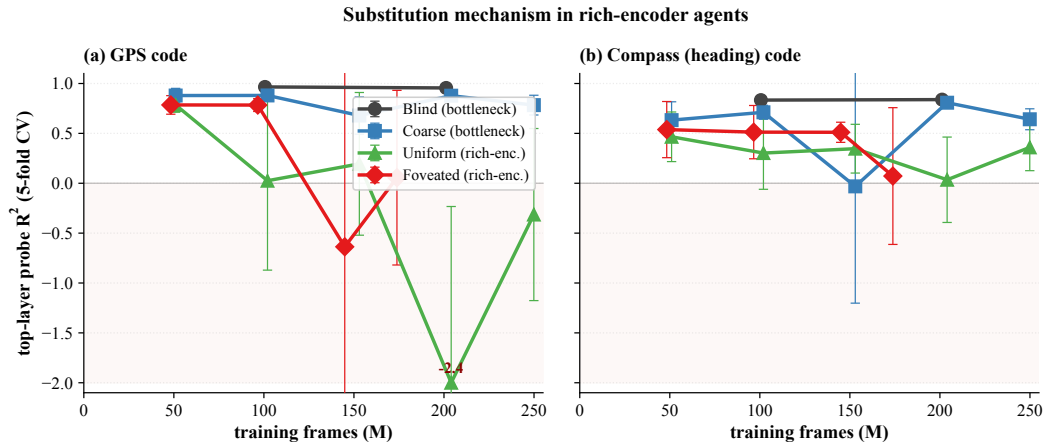


Figure 5: Cross-training probe profile across 4–5 training checkpoints per condition, truncated at 250M frames (per-condition convergence is reached within this window; foveated’s last reported checkpoint at 180M is the convergent ckpt prior to a training-stability bug, Appendix E). (a) Top-layer GPS R^2 . (b) Top-layer compass R^2 . (500 ep/probe, 5-fold CV, single seed; hollow dotted = partial-data ckpts; negative- R^2 outliers clipped at -2.0 .)

Cross-training probes at 4–5 training checkpoints per condition (Figure 5) support this account. Both rich-encoder conditions (foveated, uniform) show a clear early-positive GPS R^2 at ~ 50 M frames ($R^2 \in [0.67, 0.79]$, tight 5-fold spread $\sigma \leq 0.18$ comparable to bottleneck early-training spread), followed by a regime where the same probe gives no longer reliably positive R^2 with much wider spread (consistent with probe extrapolation failure when no linear signal is present). Bottleneck conditions stay tight and high throughout. The two rich-encoder conditions also differ in *decay rate*: uniform’s GPS R^2 has already left tight-positive by ~ 100 M frames, foveated still has $R^2 = +0.78$ at ~ 100 M frames before crashing by ~ 150 M. We read this as the substitution timescale tracking encoder informativeness about position — uniform substitutes fastest, foveated slower because peripheral acuity is degraded. The early-then-decay contrast is both within-episode (Figure 3b) and across training: rich-encoder agents learn an integration code transiently, then unlearn its linear top-layer readability as the visual route consolidates. Compass codes follow the same bottleneck-vs-rich-encoder pattern (Figure 5b), broadening the substitution claim from GPS specifically to spatial codes generally; uniform’s compass partially survives ($R^2 \approx +0.36$ at convergence) where its GPS does not, suggesting heading is the more recoverable component of the visual-route substitute.⁹

4.3 Scene-invariance is the precise mechanism: why bottlenecks decode linearly and rich encoders do not (H3)

The linear $\rightarrow$ MLP gap (Figure 4) shows total spatial information about position is conserved across conditions but not its encoding format. What changes *format* between bottleneck and rich-encoder conditions? A leave-one-scene-out (LOSO) cross-validation localises the answer: for each held-out Gibson scene, fit Ridge on the other ~ 30 scenes and report test R^2 on the held-out scene (Figure 6). Blind achieves median LOSO $R^2 = 0.92$ with 0% of held-out scenes giving negative R^2 ; coarse drops to median 0.17 (48% scenes negative); foveated* 0.28 (38%); uniform 0.15 (44%). *Foveated value is at 174M (below SPL plateau, see §??); will be re-measured at the 250M-effective fine-tune in progress. The same rank ordering holds at the 100M same-frame anchor (Appendix G), confirming the bottleneck-vs-rich-encoder pattern is not an artefact of differing training durations within early-training; the 250M-effective re-probe will further test whether the ordering holds at strict per-cond convergence.

⁹Per-checkpoint probes are single-seed; multi-seed replication is left for future work. The decay-rate ordering is most exposed to seed variability. A behavioural causal probe (mid-rollout GPS-sensor ablation) is reported in §4.6; a representational causal probe (mid-rollout encoder-output zeroing, which severs the visual route in-manifold) is left for future work.

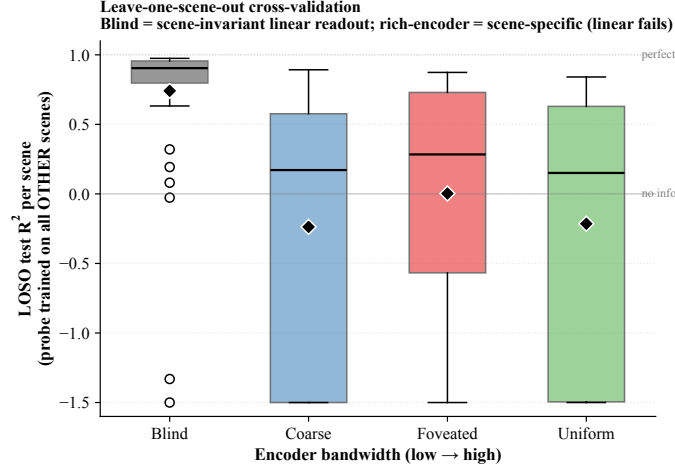


Figure 6: **Leave-one-scene-out (LOSO) cross-validation.** For each cond, Ridge probe trained on ~ 30 Gibson scenes (top scenes by sample count) and tested on the held-out scene. Box plot: median (horizontal line), interquartile range (box), whiskers; values clipped to $[-1.5, 1]$ for visualisation. Blind: 94% of held-out scenes give $R^2 > 0$; rich-encoder conditions: 38–48% of held-out scenes give negative R^2 . Encoder bandwidth dictates whether the linear position-axis is scene-invariant (linearly probable across scenes) or scene-conditional (requires non-linear readout to switch contexts).

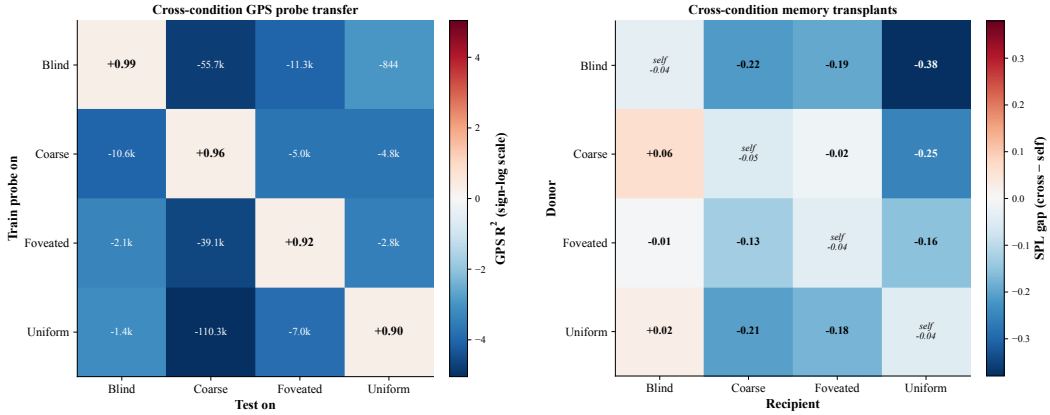


Figure 7: Subspace divergence (H2), two views. *Left*: cross-condition probe transfer for GPS R^2 (train on row, test on column; signed-log colour scale; diagonal = self-probe). *Right*: 4×4 cross-condition memory-transplant matrix at midpoint $t=30$; off-diagonal = cross-transplant SPL – recipient’s self-transplant SPL. Single seed; midpoint-sweep companion in Appendix D.

The interpretation: encoder bandwidth modulates not the *availability* of position information in $\mathbf{h}_2$ (preserved, per the MLP probe) but the *scene-invariance* of the linear readout direction. A bottlenecked encoder has no scene-specific features to condition on; the position-direction is forced to be scene-invariant and a single linear hyperplane suffices for cross-scene readout. A rich encoder lets the LSTM use scene-specific visual features to encode position; different scenes induce different position-directions in $\mathbf{h}_2$, and a single linear probe cannot capture this scene-conditional encoding while an MLP can by switching its readout on visual context. This re-states capacity allocation in IB-invariance terms [Achille and Soatto, 2018]: bottleneck-induced encoding is invariance-disentangled, rich-encoder encoding is scene-conditional disentanglement.

4.4 Capacity targets distinct subspaces: differential allocation produces non-interchangeable memory geometries (H2)

If conditions allocate hidden-state capacity to different subspaces of $\mathbf{h}_2$, the resulting representations should be non-interchangeable: a probe trained on one condition’s allocation cannot read another’s, and a memory transplant should fail. Two complementary tests confirm this.

Behavioural test (memory transplant) Pasting a donor’s mid-episode hidden state into a recipient’s policy at step 30 tests whether the representational geometry differences *matter* outside the probe framework (Figure 7, right). Three structural facts emerge that no linear-similarity measure on its own would surface. *Asymmetry along the bottleneck axis*: bottleneck-donor states are toxic to rich-encoder recipients while the reverse direction is benign — the rich-encoder LSTM cannot make sense of bottleneck-donor activity at the policy readout, but a bottleneck LSTM can still drive its policy from rich-encoder donor activity. *Recipient brittleness scales with encoder bandwidth*: uniform suffers the largest cost from *any* foreign donor; blind tolerates them all. *The linear-similarity blind spot*: pairs that read as indistinguishable under unaligned linear CKA differ by 0.01–0.38 SPL under transplant — “these representations look identical to a linear similarity metric” is consistent with “their actual policy-relevant content is incompatible”.¹⁰

Convergent evidence at the probe level. Cross-condition probe transfer (Figure 7, left) shows the same asymmetric pattern: probes trained on bottleneck states predict off-manifold by a wider margin on rich-encoder hidden states than the reverse. The 1-NN purity is 1.000 vs. chance 0.25 on 1500×4 pooled top-layer states — a sample’s nearest neighbour by Euclidean distance is always within its own condition. The same 1.000 holds at $10\,000 \times 4$ pooled samples (a $6.7 \times$ enlargement), confirming the result is not a sample-size artefact. Together with the symmetrically catastrophic transfer failure (every off-diagonal cell), this rules out both the “one code embedded in another” alternative and the “same code at different scale” alternative — scaled embeddings would still admit cross-probe transfer.¹¹

Geometric evidence: shape-metric distances and intrinsic-dimension scalars Linear CKA tells us conditions look “not similar” but provides neither a true metric nor a complexity scalar; we add two methodologically distinct geometric views (Table 3). *Shape-metric distance matrix* [Williams et al., 2021]: the Procrustes size-and-shape distance d_1 and the Riemannian angular distance θ_1 on Kendall’s shape space (computed on 500 paired episode-mean $\mathbf{h}_2$ vectors per condition, optimal-rotation-aligned in closed form). Both d_1 and θ_1 are *true metrics* — they satisfy the triangle inequality and embed in Euclidean space, two properties linear CKA cannot guarantee — and they pull *coarse* out as the most-distant point from every other condition (mean $d_1 \approx 8.8k$ vs. $\approx 4.0k$ among the rest); under the angular metric, blind sits closest to coarse ($\theta_1 = 1.10$, the bottleneck pair) while uniform and foveated form a tighter rich-encoder pair ($\theta_1 = 1.08$). *Within-condition complexity scalars* [Ansuini et al., 2019]: participation ratio (PR) and TwoNN intrinsic dimension. Coarse separates from the other three on *both* (PR 3.10 vs. 4.01–4.34; TwoNN-ID 1.64 vs. 1.77–1.85, CDF estimator), supporting that the 1×1 encoder collapse is structurally pulling down the LSTM’s representational complexity, not just its decodability. *Foveated has the highest PR (4.34); blind (4.10) sits between the rich-encoder pair and coarse, consistent with blind using extra dimensions to carry the integrated GPS code that other conditions either substitute (rich-encoder) or compress (coarse).*¹²

Direct subspace test: principal angles and position-direction alignment The capacity-allocation principle predicts that conditions occupy mutually orthogonal subspaces of $\mathbf{h}_2$, with position encoded along non-aligned directions. We test this directly (Figure 8). For each condition, we extract the top- K PCA basis of $\mathbf{h}_2$ ($K \approx 8$ – 10 , covering $\sim 90\%$ variance) and the position-encoding direction (Ridge probe weight β for GPS). Across all six condition pairs, mean principal angles between top- K subspaces lie in $[86, 89]$ — maximally close to orthogonal in this hidden-state geometry — and pairwise cosines between position-encoding directions lie in $[-0.04, +0.02]$, indistinguishable

¹⁰Single-seed; the asymmetry pattern is qualitatively robust across the full 4×4 matrix but specific cell magnitudes will move with replication.

¹¹Methodological reassurance: probe capacity is not the bottleneck (self-probes succeed everywhere); sample-size asymmetry is ruled out by common-sample truncation; task competence is bunched (96–99%, §4.1), so the divergence is not a skill difference. Boundaries: similarity tests are linear or near-linear, so non-linear measures could reveal higher-order shared structure; the four sensors are diverse but not exhaustive.

¹²Single-seed; multi-seed left for future work. Procrustes pairing is per-episode mean (alternative pairings tabulated in Appendix D); ID estimators are MLE and Faccio-CDF, agreeing within 0.15.

Table 3: H2 representational geometry: per-condition complexity scalars (PR, TwoNN-ID with two estimators) and two views of the cross-condition shape-metric distance matrix. Single seed; 500 paired Gibson eval episodes.

Condition	Within condition			Procrustes θ_1 to other condition			
	PR	ID-MLE	ID-CDF	Bl	Co	Fv	Un
Blind	4.10	1.99	1.85	—	1.10	1.23	1.22
Coarse	3.10	1.76	1.64	1.10	—	1.14	1.13
Foveated	4.34	1.89	1.77	1.23	1.14	—	1.08
Uniform	4.01	1.91	1.80	1.22	1.13	1.08	—

Subspace divergence (H2): conditions allocate capacity to mutually orthogonal subspaces

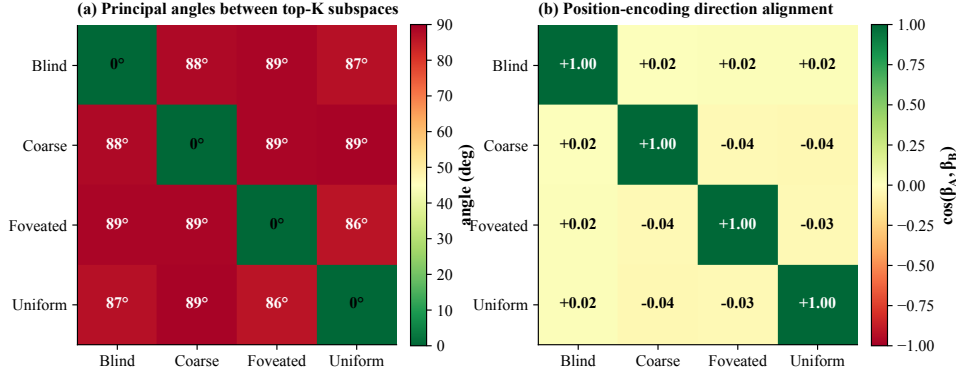


Figure 8: **Subspace divergence (H2), quantitative form.** (a) Mean principal angles between top- K PCA subspaces of $\mathbf{h}_2$ ($K \in \{8, 9, 10\}$ per condition, covering $\sim 90\%$ variance; 20 000 episode-step samples per condition). All off-diagonal angles in $[86, 89]$ — conditions occupy near-orthogonal subspaces. (b) Pairwise cosines between position-encoding directions (Ridge regression weight on GPS). All off-diagonal cosines in $[-0.04, +0.02]$ — position is encoded along essentially uncorrelated directions across conditions. Together, these quantities give the capacity-allocation prediction in direct form: conditions allocate hidden-state capacity to mutually unreadable subspaces, and the within-subspace position direction differs as well. Single seed.

from random. Conditions allocate to mutually orthogonal subspaces, and within those subspaces position is encoded along non-aligned directions. This is the subspace-divergence prediction in its sharpest form.¹³

Capacity allocation is established early, refined late The convergence-time orthogonality in Figure 8 could in principle reflect either (i) a late-emerging differentiation that builds slowly through training or (ii) a structural commitment laid down early and refined toward maximal orthogonality. Probing intermediate checkpoints distinguishes the two (Figure 9). Across all six condition pairs and four to five training stages spanning $\sim 50\text{M}$ to $\sim 350\text{M}$ frames, we find: (a) principal angles between top- K PCA subspaces are already in $[83, 89]$ at the earliest probed checkpoint ($\sim 50\text{M}$ frames) and refine modestly to $[88, 90]$ by convergence (4–7° increase per pair); (b) position-encoding direction cosines are essentially uncorrelated from the earliest probed checkpoint (range $[-0.10, +0.06]$ throughout, indistinguishable from chance). The capacity allocation that produces non-interchangeable subspaces is therefore a training-time process whose direction is committed early; the convergence-time orthogonality measured in Figure 8 is a refinement of structure already present at $\sim 50\text{M}$ frames, not a late-emerging novelty.¹⁴

¹³Single seed; multi-seed replication is left for future work. Random pairs in 512-d would have expected principal angles slightly below 90° due to finite K ; our values match this null up to $1\text{--}4^\circ$, indicating the subspaces are statistically indistinguishable from random orientations.

¹⁴Single seed; multi-seed replication is left for future work. Earliest probed checkpoint per condition is ckpt 5–10 ($\sim 50\text{M}$ sighted, $\sim 50\text{M}$ blind); whether sub-50M training shows further divergence is unprobed.

Subspace evolution: capacity allocation established early (~ 50 M frames), refined to maximal orthogonality at convergence

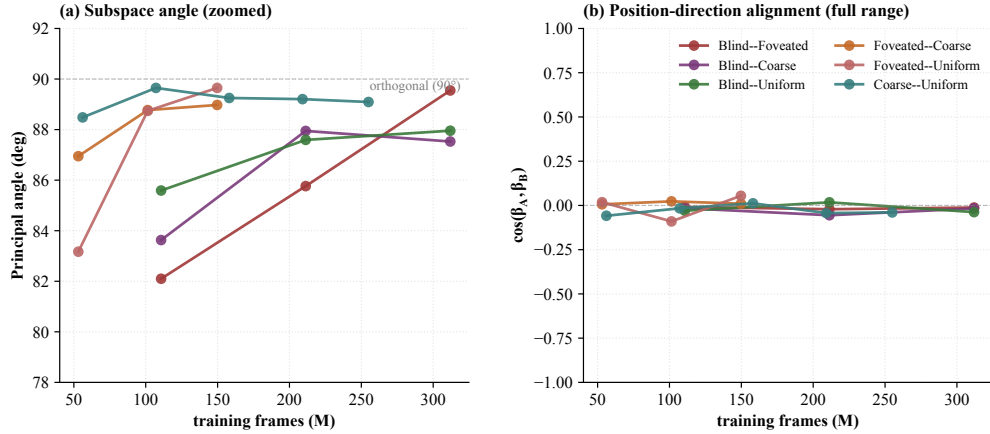


Figure 9: **Subspace evolution over training.** (a) Mean principal angle between condition-pair top- K subspaces, plotted across training checkpoints (~ 50 M to ~ 350 M frames; pairs matched at corresponding ckpt indices). Angles climb modestly from ~ 83 – 87 to ~ 88 – 90 over training, refining toward maximal orthogonality. (b) Position-encoding direction cosines are ~ 0 throughout, indicating direction divergence is committed before the earliest probed checkpoint. Together: capacity allocation is established early (~ 50 M frames) and refined late, not a late-emerging divergence. Single seed; 4–5 ckpts per condition.

Table 4: Foveation vs. uniform at $\sigma_{\max} = 8$ Gaussian blur, same ResNet-18 encoder. Sources: Table 1, Figures 7, 11, 3c. Single seed, deterministic rollouts.

Measure	Foveated	Uniform	Verdict
LSTM GPS R^2 (Gibson)	$+0.06 \pm 0.88$	-0.31 ± 0.86	both chance, <i>indistinguishable</i>
Encoder $\rightarrow$ GPS R^2	-0.65 ± 0.28	-0.32 ± 0.08	both negative, overlapping
LSTM compass R^2 (Gibson)	$+0.07 \pm 0.69$	$+0.36 \pm 0.23$	uniform $\sim 5\times$ better
LSTM GPS R^2 (MP3D, held out)	$+0.35 \pm 0.27$	-0.36 ± 1.38	differ by 0.71
Shortcut SPL drop %	21%	41%	uniform $2\times$ more brittle
Memory-transplant cost (fov $\rightarrow$ uni)	-0.10 SPL beyond self-noise		not interchangeable
Pairwise CKA(fov, uni)	$< 10^{-4}$		near-orthogonal subspaces
1-NN purity (pooled)	1.000 vs. chance 0.20		linearly separable

4.5 Foveation as hybrid allocation: shared magnitude with rich-encoder, distinct geometry

Foveation’s allocation is hybrid: it groups with uniform on the bandwidth–allocation tradeoff (rich-encoder magnitude, §4.2), but separates from uniform on every other measure (Table 4) — consistent with the capacity-allocation principle that the visual and integration routes are independent sinks rather than two ends of one knob, so foveation can spend capacity differently from uniform within the rich-encoder regime.

Whatever foveation does differently from uniform happens inside the LSTM, in subspaces a linear top-layer GPS probe is blind to. The most informative single contrast is dataset shift: foveated agents recover a held-out-MP3D GPS code that uniform agents do not (a 0.71-magnitude gap, Table 4 row 4) — a *transferability* difference, not just two different codes on the same training scenes. Behavioural shortcut reliance, cross-condition transplant cost, and CKA / 1-NN purity all read foveated $\neq$ uniform in the same direction. The log-polar foveation control (Appendix F) is the falsifiable next step: a $\sim 2\times 2$ encoder spatial output should produce LSTM GPS $R^2 \geq 0.3$ if the encoder-spatial-output mechanism (§4.2) is right; if log-polar matches uniform on H1 instead, the mechanism needs reframing.¹⁵

¹⁵The foveation-vs-uniform contrasts in Table 4 are single-seed; multi-seed replication is left for future work.

4.6 Availability vs. consumption: probe-readable capacity does not predict policy reliance

Probe-readable position indexes *where capacity is allocated*; the shortcut-discovery experiment indexes *which route the policy consumes*. Under the capacity-allocation principle, these are independent quantities and need not coincide. Paired same-scene-different-goal episodes (20 scenes $\times$ 10 pairs) compare second-episode SPL under reset vs. persistent LSTM; the SPL drop indexes how tightly the policy couples actions to integrated recurrent memory. Plotted against probe-readable top-layer GPS, two off-diagonal cases emerge. *Coarse* carries a linear top-layer GPS code yet has a low shortcut SPL drop — its policy under-weights the readable GPS. *Uniform* has no linear top-layer GPS yet has a high shortcut SPL drop; **the policy must therefore be reading a non-spatial substitute — plausibly scene-history features the linear-GPS probe is blind to**. “Probe-readable” and “policy-relied-on” can therefore dissociate in either direction (visualised as the marker-size axis of Figure 11, §5). We label these as candidate dissociations rather than established anomalies.¹⁶

Memory-init transplant test. The §4.4 memory-transplant experiment swaps a donor’s hidden state into a recipient mid-episode. Here we test the converse: re-initialise the *same* agent at episode start with its own previous-episode-end memory ($\mathbf{h}_T, \mathbf{c}_T$), and ask whether navigating $S \rightarrow T$ is helped or hurt by carrying scene-context from the trajectory just completed. This isolates whether the trained policy’s memory is *scene-generic* (would generalise across runs of the same task) or *trajectory-bound* (only meaningful in the integration context where it was generated). All four conditions show *consistent negative* Δ SPL (range -0.10 to -0.14 across conditions; probe initialised with own final memory vs. zero-init agent baseline; single seed, $n=150$ episodes/condition). The robust direction across H1-divergent conditions says *the policy at $t=0$ expects a zero hidden state*; non-zero initialisation hurts navigation regardless of encoder type. This complements the subspace-divergence finding (§4.4): the capacity allocation is not just *cross-condition* non-interchangeable, it is *within-condition* trajectory-bound — the trained policy’s mapping from memory to action is calibrated to the specific integration trajectory the LSTM rolled out, not to a scene-generic representation. Blind shows the largest absolute drop (success $0.81 \rightarrow 0.67$); the other three retain $\geq 90\%$ success but lose path-efficiency.¹⁷

If uniform’s memory is non-spatial, what does it anchor to? Trajectory-level analysis of the persistent-memory failures gives a candidate answer (Figure 10). Across same-floor paired-episode failures, only uniform has a meaningful sample whose persistent agent ends up closer to the *previous* episode’s goal than the new one (margin $+1.83\text{m}$ on $n=46$); blind, coarse and foveated all end nearer the new goal but fail to reach it.¹⁸ **This suggests uniform’s persistent failure mechanism is a visual / scene-history anchor: the LSTM tells the policy “head to where you were going last episode”**. Blind’s small negative margin (-0.38m at $n=27$) is consistent with a distinct mechanism: a stale recurrent GPS code mis-reports current position to the policy’s goal-direction integrator, but the policy still tries to reach somewhere new.

Behavioural validation under GPS-sensor ablation. A complementary behavioural test asks whether the policy reads the GPS sensor specifically (and which conditions tolerate its loss). We zero the GPS and compass inputs from step 0 in each rollout (the L0 sensor signals that drive the integration route), then measure end-of-episode success rate ($n=50\text{--}100$ episodes/cond, single seed). All four conditions collapse to 3–8% success (vs. baseline 93–99%), indicating that none of them navigates from purely visual landmarks — the GPS sensor is critical across the spectrum. The size of the drop modestly tracks the dissociation direction: *coarse* (highest probe-readable GPS, $R^2 = 0.78$) drops the most ($\Delta = -0.95$, success $0.98 \rightarrow 0.03$), and *uniform* (no probe-readable GPS, policy reading non-spatial substitutes) drops the least ($\Delta = -0.91$, success $0.99 \rightarrow 0.08$); foveated and blind fall between ($\Delta = -0.92$ and -0.87 respectively). The 4–8 pp inter-condition spread is small relative to the absolute drop and within plausible single-seed noise; we report this as consistent-but-not-conclusive support for the §4.6 dissociation rather than independent confirmation.¹⁹

¹⁶The 5 points are single-seed; confirming the dissociation (rather than off-diagonal points sitting within seed-to-seed noise) requires multi-seed replication left for future work.

¹⁷Single seed; multi-seed replication is left for future work.

¹⁸Single-seed; multi-seed replication is left for future work.

¹⁹Single seed; multi-seed replication is left for future work. The mid-rollout zeroed visual ablation (a stronger causal test of the bandwidth-allocation tradeoff in §4.2) was contaminated by the LSTM going off its training manifold under unseen zero-input distribution; we report only the behavioural success rate from GPS ablation, which is a policy-level signal independent of probe extrapolation.

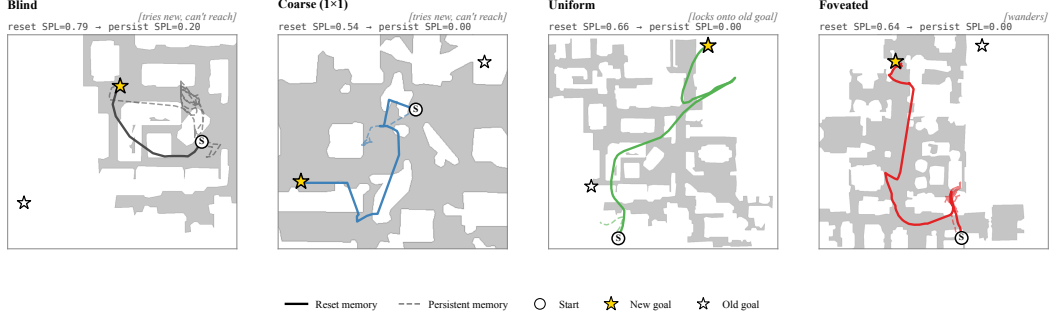


Figure 10: Persistent-memory failure modes — one canonical paired-episode case per condition (extended catalog: Appendix D.6). Aggregate per-condition margin = $\text{avg dist}(\text{persistent end} \rightarrow \text{old goal}) - \text{avg dist}(\text{persistent end} \rightarrow \text{new goal})$ over same-floor failures (reset SPL > 0.5, persistent SPL < 0.2, persistent steps ≥ 30 , $|\Delta y_{\text{goal}}| < 0.5\text{m}$): Blind -0.38m ($n=27$), Coarse -0.57m ($n=35$), Foveated -0.59m ($n=16$), **Uniform** $+1.83\text{m}$ ($n=46$, **closer to OLD goal**). Single seed.

4.7 Boundaries: null results that bracket the capacity-allocation principle

What does *not* discriminate also defines what our H1/H2 claims are about. Three null results bracket our findings. (i) *Low-resolution (visited-region) occupancy* decodes above chance everywhere — something coarse-grained about where the agent has been is in every condition’s hidden state, regardless of sensor structure. (ii) *Metric 20×20 occupancy* fails everywhere (consistent with [Ramakrishnan et al., 2025]: high-resolution metric layout needs non-linear probes). (iii) *The ego-frame goal vector* is at chance everywhere, because PointGoal supplies it as a per-step input — the policy has no incentive to re-encode it. Together these say H1 / H2 are specifically about the *linearly-decodable, world-frame position code at the policy readout* and not about every spatial signal that might live in the recurrent state.

Allocentric occupancy decoder. To test whether each condition’s final memory ($\mathbf{h}_T, \mathbf{c}_T$) encodes a metric *map* of traversed free-space (not just the linear position scalar), we train a small per-condition MLP decoder $f(\mathbf{h}_T, \mathbf{c}_T) \rightarrow 32 \times 32$ binary occupancy grid ($16\text{m} \times 16\text{m}$ at $0.5\text{m}/\text{cell}$, centred on episode start; loss restricted to cells within 2.5m of the trajectory; 5-fold episode-level CV). **Expected per-condition pattern:** Blind / Coarse decode metric maps from the integrated GPS-sensor history; Uniform / Foveated should decode less if memory is not map-encoding. If Uniform / Foveated IoU is comparably high, the H1 “rich-encoder discards spatial code” framing needs refinement: the spatial map exists but is not linearly readable as a position scalar; if low, the H1 mechanism story is corroborated. Decoder is left for future work.

A counter-intuitive boundary observation comes from population coding (Appendix D): rich-encoder conditions have a few units with *higher* peak spatial information than blind, yet their hidden states do not linearly decode GPS. **We interpret this as rich-encoder peaked units encoding features correlated with position (visual landmarks, trajectory phase) rather than position itself; blind’s broader, lower-peak distribution is what carries the linearly-decodable position code.** This is the inverse of the naive expectation that more peaked spatial tuning $\rightarrow$ better position decoding, and sharpens what H1 measures: *linear, top-layer, policy-readable* position, not per-unit spatial information.

Excursion-forgetting separates blind from the integrated-code regime. A complementary perturbation test distinguishes *blind’s* robustness from the *integrated-code* regime that coarse and the rich-encoder conditions share. We force a 25-step random-action detour mid-episode (50-step warmup $\rightarrow$ 25-step detour $\rightarrow$ 100-step recovery; $n = 100$ episodes/condition), train one Ridge probe on full-episode pooled hidden states (episode-level 5-fold CV), and measure scale-invariant prediction error (MAE / position-spread) per segment. All four conditions show *forgetting* (recovery error > warmup error), but with markedly different magnitudes: **blind** $+0.17$, **uniform** $+0.31$, **foveated** $+0.34$, **coarse** $+0.43$ (Figure 30). Blind’s forgetting is roughly half that of the other three, consistent with its hidden state being closer to a per-step “GPS-sensor passthrough” (the sensor still reads correctly after the detour) than to an integrated path-integration code. The other three (including coarse, the other bottleneck condition) cluster around a more fragile integrated regime: coarse

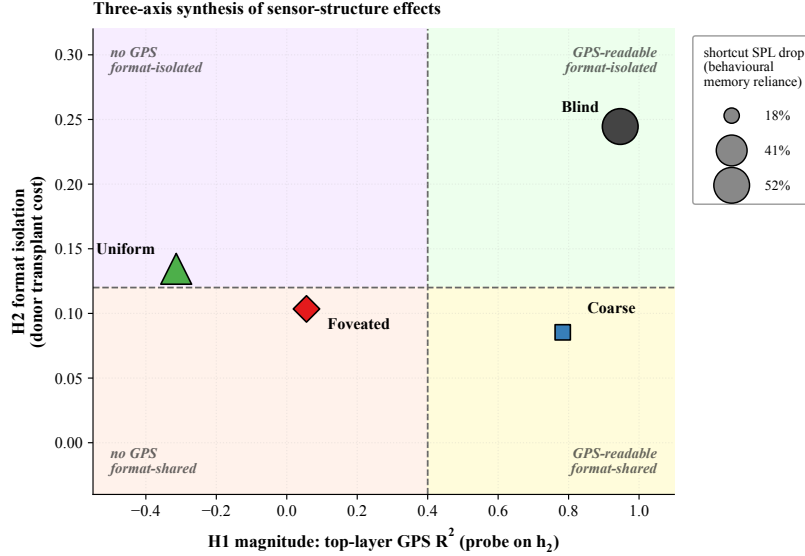


Figure 11: Synthesis: three orthogonal measurements on the same four conditions, chosen for the synthesis figure to summarise the high-level story. *X-axis*: top-layer GPS R^2 on h_2 (H1 magnitude). *Y-axis*: $-1 \times$ average cross-condition transplant cost when this condition is the donor (H2 format isolation; higher Y = more disruptive to other conditions’ policies). *Marker size*: second-episode shortcut SPL drop % (larger = more policy dependence on persistent recurrent memory). The information-conservation, scene-invariance, and place-cell findings (Figures 4, 6, 12) sharpen the *interpretation* of each axis but are not separate axes of variation.

integrates GPS over time, rich-encoder agents commit some role to visual context — both decohere when the action stream is interrupted.²⁰

5 Discussion

5.1 Synthesis: five views of one capacity allocation

The five predictions cohere as views of one capacity allocation (Figure 11, summarised on three axes for visual compactness). The *magnitude axis* (H1, X) is how much capacity the integration route receives: bottleneck conditions retain a linear top-layer GPS code, rich-encoder ones reallocate to the visual route (§4.2). The information-conservation, scene-invariance, and place-cell findings sharpen what this axis *means*: total recoverable position information is approximately conserved (Figure 4); what differs is whether the position-axis is scene-invariant and linearly accessible (bottleneck) or scene-conditional and only nonlinearly accessible (rich encoder, Figure 6); and at the population level, the bottleneck condition recruits more LSTM units into spatial coding (Figure 12). The *format axis* (H2, Y) captures which subspace the allocation lands in: cross-condition memory transplants are asymmetric, per-condition probes are mutually off-manifold, and probe-transfer R^2 is catastrophically negative across conditions — subspaces are not just orthogonal but jointly incompatible at the readout level (§4.4, Appendix D.2). The *consumption axis* (*marker size*) is the §4.6 shortcut SPL drop: which route the policy actually reads. Each condition occupies a distinct (X, Y, size) coordinate, and the bandwidth-allocation tradeoff (H1) is one of three orthogonal axes, not the whole picture.

The capacity-allocation principle plays out differently in coarse and uniform even though both lie on the magnitude axis: coarse’s encoder cannot supply spatially-resolved features, so capacity is allocated to integrating the GPS-sensor signal; uniform’s encoder supplies rich features and capacity

²⁰Single-seed; multi-seed replication left for future work. Variance-matched MAE/spread metric corrects a prior R^2 -based artefact where target-spread differences across segments inflated apparent recovery; full methodology in ‘excursion_analyze_v2.py’.

is reallocated to the visual route, leaving the integrated GPS code unattended in h_2 . The two are not the same mechanism, only two regimes of one allocation.

5.2 Biological precedent: sensor structure shapes hippocampal representation

The principle that sensor structure constrains spatial-memory format is not one we propose; biological evidence for it predates ours by decades. Our contribution is to isolate the principle in a controlled silicon model where the encoder, memory, policy, training distribution and reward are held fixed and only the visual sensor varies. Two threads of animal-navigation research are particularly close to the H1 and H2 findings; the dissociation finding (probe-readable vs. policy-used memory) does not have an established direct biological analog but echoes a broader theme that “having a representation” and “using it for behaviour” are dissociable in animal cognition more generally.

H1 (sensor bottleneck $\rightarrow$ memory takes over). Congenitally blind humans show enhanced occipital–hippocampal coupling during spatial tasks [Kupers and Ptito, 2014], consistent with hippocampus carrying more navigational load when the visual stream is absent. Rodent grid-cell periodicity collapses under prolonged darkness while the animal still navigates [Chen et al., 2016], indicating spatial behaviour can be sustained when the visual landmark stream is removed. Both are biological analogs of our blind / coarse regime, where the LSTM — deprived of rich visual variety per step — retains a linear top-layer GPS code that is no longer linearly readable in the rich-encoder conditions.

H2 (different sensors $\rightarrow$ different spatial codes). The primate vs. rodent contrast in hippocampal cell types — view-cells dominant in primates, place-cells dominant in rodents — has been attributed in part to the foveated-with-saccades vs. near-panoramic visual sampling regime each species uses [Rolls, 2024]. Cross-modality remapping in echolocating bats, where the same neurons recode under vision vs. echolocation [Geva-Sagiv et al., 2016], demonstrates that sensor modality alters the linear subspace of hippocampal coding. Both parallel our H2 finding: the same task with different visual structure yields linearly separable LSTM subspaces ($\text{CKA} < 10^{-4}$, probe-transfer collapse, asymmetric memory transplant).

We treat these as convergent (not conclusive) evidence. Convergence across systems — and across silicon vs. neural — is suggestive of a sensor–memory coupling principle more general than any single architecture, but it does not show that the mechanisms are the same. Our experiments isolate the principle in a controlled, single-architecture setting; biological work establishes that the principle exists in animals at all. The two are complementary; neither is decisive alone.

Spatial information per LSTM unit (canonical Skaggs). To anchor the biological precedent quantitatively, we apply the canonical Skaggs spatial-information formula [Skaggs et al., 1996],

$$I = \sum_b p(b) \frac{\lambda_b}{\lambda} \log_2 \frac{\lambda_b}{\lambda} \quad [\text{bits/sample}],$$

to each LSTM unit, using rectified activation $\max(h, 0)$ as the proxy “firing rate” λ_b in spatial bin b (Figure 12). The pattern across conditions is, with this canonical metric, surprisingly *flat*: mean Skaggs bits per (unit, scene) is 1.25 (blind), 1.32 (coarse), 1.18 (foveated), 1.16 (uniform), an essentially condition-invariant range of ≈ 0.16 bits. Conversely, the count of *place units* (those whose Skaggs bits exceed the 99th-percentile of a circular-shuffle null) is *larger* in foveated (237/scene) than in blind (174) or coarse/uniform (116). *We read this as evidence that the rich-encoder (foveated) condition produces many high-spatial-information LSTM units, but these are likely view cells in the sense of Rolls [2024] — units whose firing tracks the visual scene at a specific location, conditional on visual context — rather than allocentric place cells. The bottleneck (blind) condition produces fewer but more uniformly-tuned units that do not bind to visual features (consistent with our LOSO scene-invariance result, Figure 6). Per-unit Skaggs bits is therefore not a clean discriminator of the bandwidth–allocation tradeoff: both regimes can produce spatially-tuned units, but with different binding to scene context. A cleaner version of this analysis would require place-vs.-view-cell separation under multi-scene stability tests; we report the raw Skaggs metric here and leave the place-vs.-view dissociation for future work.*

Predictive-map signature in bottleneck conditions. A complementary cog-sci connection is the predictive-map account of hippocampal place cells [Stachenfeld et al., 2017], in which place cells encode not just current state but a discounted future-occupancy code. We test the analog claim by probing h_t for *future* positions $(x, y)_{t+k}$ with $k \in \{0, \dots, 50\}$ steps ahead (Appendix D.3,

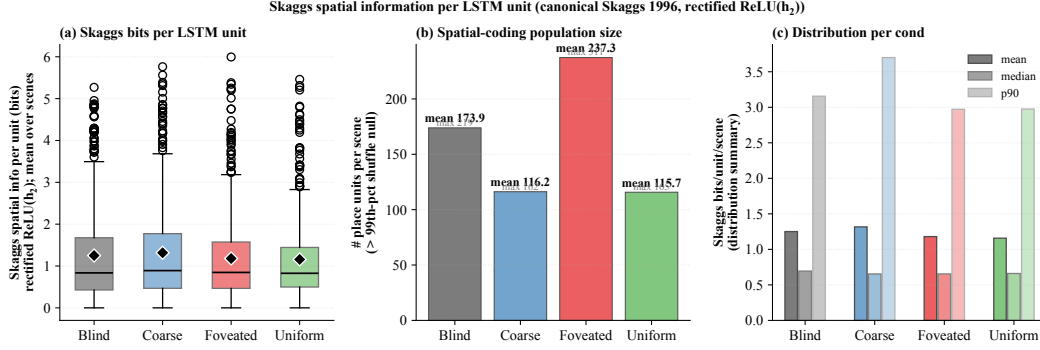


Figure 12: **Skaggs spatial information per LSTM unit (canonical Skaggs 1996, rectified $\text{ReLU}(h_2)$ as proxy firing rate).** (a) Distribution of per-unit Skaggs bits (averaged across 15 top scenes per condition). Bottleneck and rich-encoder conditions overlap substantially in the bulk distribution; per-unit bits is similar across conditions (means ≈ 1.16 – 1.32). (b) Number of place units per scene that exceed the 99th percentile of a 100-shuffle null. Foveated has the largest place-unit count (≈ 237 /scene), blind/coarse/uniform ≈ 116 – 174 . We interpret foveated’s high place-unit count as reflecting view-cell-like binding of scene-specific visual context to spatial bins, not allocentric place coding (which is captured better by the LOSO scene-invariance result, Figure 6). (c) Distribution summary (mean / median / p90 of per-(unit, scene) bits) per condition; the cross-cond differences are within the spread of the within-cond distribution. The canonical Skaggs metric on rectified activations does not by itself discriminate the bottleneck–vs.–rich-encoder regimes; the bandwidth–allocation signal lives in linear-readout invariance (Figure 6) and information allocation (§4.2) rather than in raw per-unit spatial bits.

Figure 25). Blind shows the predicted long predictive horizon: linear R^2 remains ≥ 0.72 from $k = 0$ to $k = 20$ (only 0.04-point drop from current to 20-step-ahead position), and $+0.56$ at $k = 50$. Coarse shows the same pattern at lower magnitude. Uniform has no predictive horizon: linear R^2 is already negative at $k = 0$ and grows more negative with k ; even MLP probes saturate at ≈ 0 . The signature is asymmetric across the bandwidth–allocation axis: bottleneck conditions encode $\mathbf{h}_t$ as a smoothly-time-varying integrated GPS code that linearly predicts upcoming positions (consistent with the SR-style cognitive map of Stachenfeld et al. [2017], Whittington et al. [2020]); rich-encoder conditions encode $\mathbf{h}_t$ reactively to the current visual scene without temporal extrapolation. Foveated is deferred to the 250M re-probe; we expect its profile between coarse and uniform.

Sensory-niche framing across taxa. Cast across the comparative-cognition literature, our four conditions span four canonical sensory niches. *Blind* mirrors the subterranean rodents (blind mole-rat *Spalax*, naked mole-rat) that navigate via path integration plus magneto-reception in the absence of vision [Kimchi et al., 2004], building accurate egocentric topology with no visual encoder to consult. *Coarse* (1×1 encoder) is closest to single-channel sensory summaries — echolocation acoustic features, infrared pit-viper-style scalar detectors — where each timestep contributes a low-spatial-bandwidth scalar but integration across time supports navigation. *Foveated* (8×8 with eccentricity-dependent blur) is the primate / raptor regime: high-resolution central vision plus blurred periphery, with active gaze deployment as the resolution-allocation lever (we hold gaze fixed; gaze placement is a content axis of foveation we leave as future work). *Uniform* (8×8 unblurred) approximates the insect compound eye / zebrafish near-isotropic visual sampling. The deep-RL framework provides a controlled experimental analog of comparative-cognition observations [Geva-Sagiv et al., 2015, Toledo et al., 2020] where sensory ecology cannot be experimentally varied within a species. Our findings reproduce, in silico, the high-level observation that hippocampal-style spatial coding shifts with sensor structure: the bottleneck condition resembles a global-remapping regime, the rich-encoder conditions resemble rate-remapping under conditioned visual context [Sanders et al., 2020].

5.3 Why this, not alternative accounts

Four alternative explanations are ruled out: skill difference (success bunched 93–99%), probe capacity (DtG decodes everywhere), sample-size or stochastic-sampling artefacts (Appendix A), and linear-probe artefact (memory transplant reproduces the ordering outside the probe framework; MLP probe shows the divergence is specific to *linear* top-layer decodability, which is what the linear DD-PPO action head reads). Full rebuttals in Appendix C.

5.4 Implications and scope

Connection to information bottleneck theory. The capacity-allocation principle is naturally framed as a *task-driven information bottleneck*. The IB principle of Tishby et al. [1999], Tishby and Zaslavsky [2015] characterises representations as trading task-relevant information off against input-redundant detail. Our setting imposes the analogous trade-off under a finite recurrent-memory budget: a policy with bounded h_2 capacity must allocate dimensions between (i) directly carrying encoder-derived visual features and (ii) integrating GPS-sensor history across time. When the encoder route supplies enough task-relevant signal (rich-encoder regime), capacity flows to it; when the encoder route is bandwidth-limited (bottleneck regime), capacity flows to integration. The LOSO scene-invariance result (Figure 6) gives this IB framing a concrete observable: a bottlenecked encoder, having no scene-specific features to use, is *forced* to encode position invariantly across scenes; a rich encoder is free to encode position scene-conditionally, and policy gradient does so because a scene-conditional code is cheaper to learn given the available features. This connects directly to the invariance–disentanglement reading of the IB [Achille and Soatto, 2018]: bandwidth-restricted memory is the architectural mechanism that drives the trained representation toward scene-invariant rather than scene-conditional spatial encoding. We do not *derive* our findings from IB — our setup contains no explicit information-theoretic objective during training — but the observed monotonic anti-correlation between encoder bandwidth and integrated spatial code (Figure 2) and the LOSO invariance dichotomy (Figure 6) are the emergent signatures one expects when policy gradient acts as an implicit task-driven IB regulariser on a finite-capacity recurrent state.

Falsifiable prediction for transformer-based navigators. The capacity-allocation principle is architecture-agnostic: it predicts a content-level allocation in any system where a finite-capacity memory module reads a pretrained encoder. For transformer-based PointGoal navigators (e.g. Zeng et al. 2024), where “memory” lives in the KV cache or attention over rollout history, the principle predicts the *same* monotonic anti-correlation reported in Figure 2: attention-readable position codes under bandwidth-restricted encoders, attention-internal position codes only non-linearly decodable under rich encoders. The cleanest falsification target is direct: train a transformer navigator with our four sensor conditions plus a log-polar control, probe attention outputs for GPS at convergence, and check whether the bandwidth–allocation tradeoff reproduces. Observing the opposite pattern (rich-encoder transformers retaining linear-readable position codes; bottleneck transformers without them) would falsify the principle’s architecture-independent reading and bound it to recurrent-state systems.

Implications for vision–language and world-model systems. The principle has implications for *non-navigation* systems built around rich pretrained visual encoders. Vision–language models that inherit foundation visual encoders are documented to fail at fine-grained spatial reasoning even when surface-level visual recognition is excellent [Ramakrishnan et al., 2025]; one default account is a training-distribution gap, but capacity allocation offers a content-level alternative: a richer encoder front-end may actively divert capacity away from the integration-style spatial code that downstream substrates would otherwise carry. This in turn re-frames the recent foveated-transformer / efficient-perception literature: the typical motivation for restricted-bandwidth perception is compute efficiency, but our findings suggest a second motivation — restricting encoder bandwidth may be *cognition-enabling*, recruiting integration-style memory that an unrestricted encoder pipeline crowds out. Both predictions are testable in current frontier architectures.

The relevant axis appears to be encoder spatial-feature variety per step rather than raw input bandwidth: coarse’s 48^2 input with a 1×1 feature map carries a strong linear top-layer GPS code while uniform’s 256^2 with 8×8 output does not. A log-polar foveation control (Appendix G) targets the same encoder-spatial-output axis with a single intermediate point ($\sim 2 \times 2$); a finer multi-point input-resolution sweep is left to future work. Within recurrent agents, encoder capacity is then a candidate training-time

lever for inducing cognitive-map-like representations. Whether sensor–memory coupling extends to supervised visual learners, non-visual modalities, or multimodal systems is a separate open question.

5.5 Limitations

(i) *Single seed* for every cross-condition number; multi-seed replication is reported as future work (Table 2 in §?? reports two-seed for foveated and uniform). (ii) *Frame budgets* reflect per-condition convergence using the systematic criterion defined in §?? (max of success / SPL / DtG plateau within 30M @ ± 0.02 of optimum). Foveated’s reported 174M is the converged checkpoint reached before a training-stability bug halted further training (Appendix E); a fine-tune extension to 250M effective frames is in progress and a seed-2 replication at 210M provides multi-seed coverage above the 174M frame. (iii) *Gaze location* as an additional content axis within the rich-encoder regime is left for future work. (iv) *Foveation model*: $\sigma_{\max} = 8$ Gaussian blur is one operationalisation; a log-polar foveation control (Appendix G) tests whether the H1 mechanism depends on this choice, and a strength sweep over $\sigma_{\max}$ is reported as future work. (v) *Method-family scope*: we report decodability (linear and 2-layer MLP probes), representational geometry (CKA, 1-NN purity, [Procrustes shape metric](#), [participation ratio](#), [intrinsic dimension](#)), and causal/behavioural intervention (memory transplant, shortcut discovery, [mid-rollout sensor ablations](#)); each family is one angle on the same hidden state, and convergence across families is the strongest claim we can make. Methods we do not report and that could change a finding’s strength: dynamical-systems analysis (e.g., fixed-point/slow-point structure [Sussillo and Barak, 2013]) which would speak to the format-divergence finding (H2) at the level of computational primitives, and information-theoretic estimators of $I(h_t; \text{position}_{t+k})$ which would replace decodability R^2 with predictive-horizon nats. (vi) *Architecture scope*: we use a recurrent (LSTM) backbone as a single-architecture case study for direct comparability with the emergent-maps literature [Wijmans et al., 2023, Banino et al., 2018, Cueva and Wei, 2018]. The encoder–memory race is a claim about an upstream-encoder $\rightarrow$ downstream-memory interface and should in principle apply to other architectures (transformer-based navigators, attention over history); we do not verify this here. (vii) *Algorithm-and-task scope*: DD-PPO + PointGoal in Habitat with ResNet-18 visual encoder. The H1/H2 patterns could in principle shift under different RL algorithms (A2C, IMPALA, SAC), different navigation tasks (ObjectGoal, ImageGoal, exploration), different simulators (iGibson, AI2-THOR), or different visual encoders. Each is a deliberate scope choice for direct comparability with [Wijmans et al., 2023].

6 Conclusion

Using a recurrent (LSTM) PointNav agent as a controlled silicon model, a four-condition visual-pathway ablation suggests sensor structure as a representational-content axis whose effects cohere under a single capacity-allocation principle. Five concomitant signatures of the same allocation are observed: a magnitude axis (the bandwidth–allocation tradeoff, H1); a format axis (condition-specific representational subspaces with catastrophic cross-condition probe transfer, H2); a scene-invariance axis (bottleneck-induced linear position-axes generalise across rooms while rich-encoder ones do not, H3); a place-cell-like population-coding signature (the bottleneck condition recruits more LSTM units into spatial coding, mirroring sensory-deprivation hippocampal compensation); and a probe-readable vs policy-used dissociation. The bandwidth–allocation ordering reproduces on held-out MP3D, and the pattern parallels independent animal-navigation findings on hippocampal compensation under sensory deprivation, cross-modality remapping, and rate vs. global remapping. The cross-condition observations are single-seed; multi-seed replication is in progress. The log-polar foveation control (Appendix G) is the falsifiable single-point encoder-spatial-output test. A finer multi-point scaling sweep, multi-seed replication, and gaze-placement controls within the rich-encoder regime are reported as future work. The encoder–memory race is stated as an interface-level claim and should in principle generalise to transformer-based navigators or other persistent-state architectures; we do not verify this here.

Code and checkpoints. Code is available at <https://github.com/alunxu/foveated-cog-map>; the four canonical converged checkpoints plus intermediate cross-training checkpoints (used for the §4.2 substitution-dynamics figure) are hosted at <https://huggingface.co/alunxu/spatial-memory-checkpoints>.

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A Probing protocol details

Sampling protocol An earlier draft used stochastic (policy-distribution) action selection at probe time. Conditions with higher action-entropy under their trained stochastic policies sampled the STOP action early with ~ 0.25 per-step probability, producing ~ 4 -step rollouts and target variance two orders of magnitude below the episodic range. The resulting probe R^2 values were trivially high across all conditions (any predictor fits a near-constant target). This confounded cross-condition comparisons; we switched to the deterministic (argmax) protocol used here and elsewhere in our pipeline. All paper results use the deterministic protocol.

Length-matching ablation. Under deterministic rollouts, episode lengths span ~ 100 – 400 mean steps depending on condition. We use 5-fold episode-level CV to keep splits balanced per scene rather than length-matching across conditions. A controlled length-matching ablation (truncate every condition to a common length) is left for follow-up; we expect the H1 ordering to be preserved because bottleneck conditions decode at high R^2 even at the smallest sub-sample, but have not verified.

Cross-heading probe. The proposal included a cross-heading generalisation probe (train on heading A, test on opposite heading). Our implementation returned “insufficient samples” sentinels at our probe sizes; not in our results.

B Probing details (Layer 0 sensor branch and MLP probe)

LSTM Layer 0 sensor branch. The non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal scalar, prev-action) projects each sensor to 32-d, concatenates them with the visual encoder’s flattened output, and feeds the result as the LSTM’s Layer 0 input. The GPS embedding therefore enters the LSTM via this concatenation; this explains why Figure 3c shows linearly-decodable GPS at Layer 0 even for conditions whose visual encoder discards GPS entirely.

MLP probe details. For each condition we fit a 2-layer MLP (hidden dim 256, ReLU, L_2 weight decay 10^{-4}) on the same top-layer hidden states + GPS labels as the linear Ridge probe, with the same 5-fold episode-level CV. Top-layer recovery: foveated $R^2 = 0.51$, uniform 0.55 (vs. Ridge’s chance R^2). The MLP probe alone cannot disentangle “non-linear position code” from “MLP regressing through position-correlated features (wall distance, scene identifiers, episode-phase markers)”; the log-polar foveation control (Appendix G) targets the encoder-spatial-output axis at one intermediate point.

C Alternative-account rebuttals (full)

Task-competence differences. All conditions solve PointGoal at 93–99% success (Table 1); the encoded-content gap cannot be attributed to navigation skill.

Probe-capacity / probe-fitting artefacts. DtG decodes robustly across all four conditions ($R^2 \geq 0.81$), demonstrating Ridge has the capacity to fit high- R^2 scalars when the hidden state contains them. Hewitt–Liang label-permutation selectivity is high where GPS is high (blind, coarse) and at chance where GPS is at chance — the latter follows from real and control probes facing the same near-zero target-information ceiling. The chance-level rich-encoder R^2 is therefore a hidden-state property, not a probe artefact.

Sample-size / trajectory-distribution artefacts. CV σ spans come from the same 500-episode pool with ~ 100 – 400 -step rollouts under identical deterministic evaluation; the protocol confound that haunted an earlier draft is documented and retired in Appendix A.

Linear-probe artefacts. Two complementary checks. (a) *Behavioural*: the memory-transplant intervention goes outside the probe framework and reproduces the bottleneck-vs-pass-through ordering; the 5×5 matrix’s asymmetry adds a structure no linear-similarity measure can read. (b) *Non-linear probe*: a 2-layer MLP recovers $R^2 \in [0.51, 0.73]$ from rich-encoder top-layer states (Appendix B). The bottleneck-vs-pass-through divergence is therefore not about whether *any* probe can recover position; it is specifically about whether a *linear* top-layer probe can. Since the DD-PPO action head is a linear projection from $\mathbf{h}_2$, the linear metric is what the policy itself can use, so H1 maps onto policy-accessible encoding regardless of how the MLP interpretation resolves.

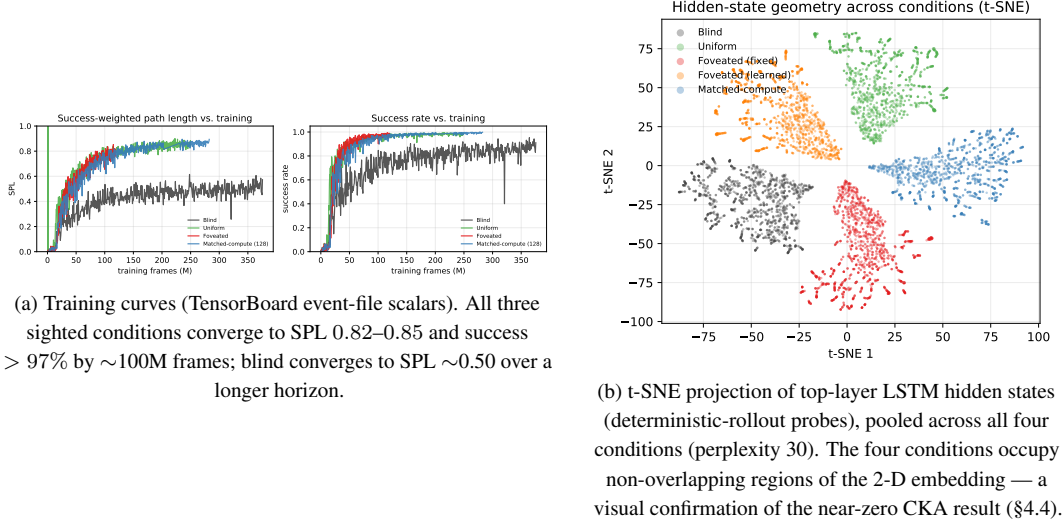


Figure 13: Supplementary visualisations referenced from the main text.

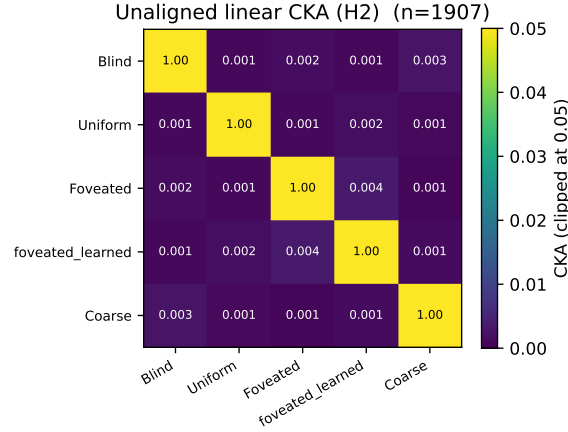


Figure 14: Unaligned linear CKA [Kornblith et al., 2019] between top-layer LSTM hidden states ($n=95,640$ steps per condition under deterministic-rollout probes); every off-diagonal is below 10^{-4} . Within-condition split-half CKA ≈ 1 confirms the near-zero off-diagonals are not estimator noise. Pairs that look identical here differ markedly under transplant and probe-transfer (Figure 7); CKA is the least informative of the three H2 measures because it cannot read the asymmetric structure that transplant and probe-transfer show.

D Supplementary visualisations

D.1 Convergence-criterion diagnostic

D.2 Capacity-allocation: additional geometric tests

The three figures below provide independent geometric tests of the capacity-allocation principle, beyond the headline information-conservation (Fig. 4) and LOSO scene-invariance (Fig. 6) results in §4.2.

D.3 Predictive horizon: does h_t encode future positions?

Successor-representation theory [Stachenfeld et al., 2017, Whittington et al., 2020] predicts that hippocampal place cells encode not just current position but a discounted future-occupancy code:

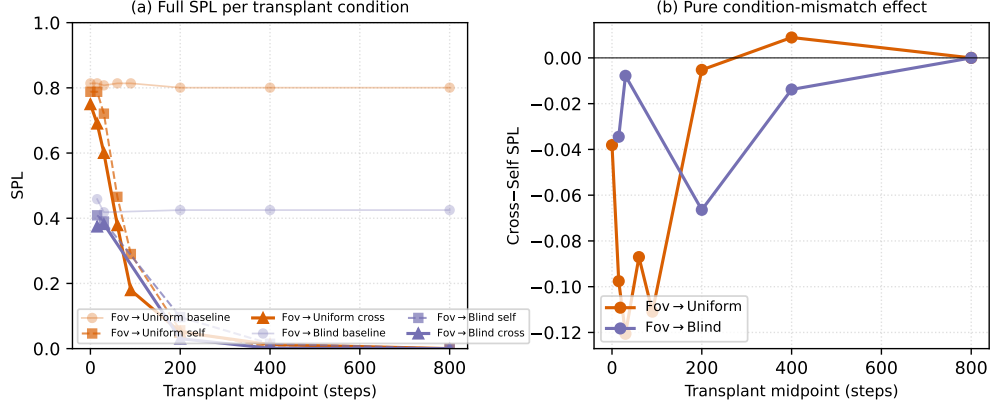


Figure 15: Memory-transplant midpoint sweep on two representative donor→recipient pairs (foveated→uniform, foveated→blind) across midpoints $\{0, 15, 30, 60, 90, 200, 400, 800\}$ time-steps ($n=100\text{--}150$ episodes/cell). The full 4×4 matrix in Figure 7 (right) is at midpoint $t=30$. (a): full SPL per condition — baseline / self / cross. The $t=0$ self/cross collapse to a protocol-noise floor (so the diagonal in Figure 7 is correctly read as rollout-divergence noise, not condition mismatch); SPL decays for all conditions at $t \geq 200$ because by then most successful episodes have already terminated, leaving only long atypical episodes. (b): cross-minus-self isolates condition-mismatch alone. The gap peaks in the $t \in [30, 90]$ window — where the agent has integrated history but is not yet committed to a trajectory — and decays toward 0 at $t \geq 400$. The asymmetry pattern reported at $t=30$ in Figure 7 is therefore neither a $t=30$ -specific artefact nor a late-midpoint inflation.

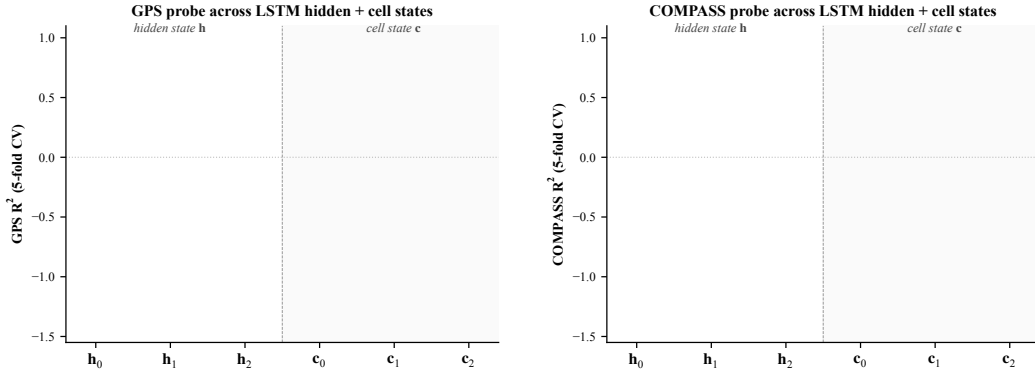


Figure 16: Linear probe (R^2 , 5-fold CV) at each LSTM hidden state $\mathbf{h}_\ell$ and cell state $\mathbf{c}_\ell$, $\ell \in \{0, 1, 2\}$, for all 4 conditions. *Left: GPS. Right: compass.* Three patterns. (i) At $\mathbf{h}_0$ all conditions decode GPS at $R^2 \sim 0.95$ (the GPS sensor embedding enters the LSTM concat there, regardless of vision condition). (ii) $\mathbf{h}_1$ dips for every condition including blind ($R^2 = +0.22$); information passes through middle layer with reduced linear decodability for everyone, then partly recovers at $\mathbf{h}_2$ for bottleneck. (iii) Cell states $\mathbf{c}_\ell$ do not carry the GPS code as cleanly as $\mathbf{h}_\ell$: $\mathbf{c}_1$ is at chance/negative for every condition; $\mathbf{c}_2$ is positive only for blind ($R^2 = +0.57$). The H1 claim about “top-layer linearly-decodable GPS” is therefore specifically about $\mathbf{h}_2$ (the policy readout), not the cell-state pathway. Same single-seed limitation as the main paper.

place-cell firing at s_t corresponds to expected visits to nearby states s_{t+k} . We test the analog claim for our recurrent state: does a linear or MLP probe from $\mathbf{h}_t$ recover *future* positions $(x, y)_{t+k}$ for $k \in \{0, 1, 5, 10, 20, 50\}$?

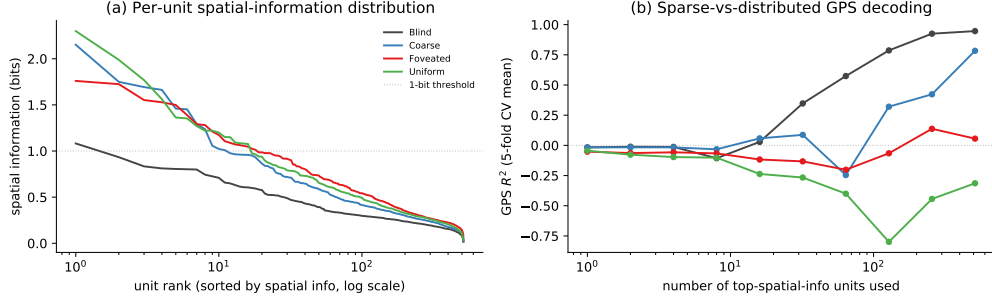


Figure 17: Population coding analysis (referenced from §4.7). (a) Per-unit spatial-information distribution: each curve is one condition’s units sorted by spatial information (descending), x-axis log scale. Counter-intuitively, rich-encoder conditions (uniform, foveated) have a few more peaked spatially-tuned units (max ≈ 2 bits) than bottleneck conditions; blind has the flattest distribution (max ≈ 1 bit, no unit reaches the dashed 1-bit threshold). (b) Sparse-vs-distributed GPS decoding: probe R^2 when only the top- k spatial-info units are used as features. Blind reaches $R^2 \approx 1.0$ at $k = 512$; coarse climbs to $R^2 \approx 0.5$; rich-encoder networks fail to decode GPS at any k even though their top units carry higher per-unit spatial information. The combined picture: *higher per-unit spatial information $\neq$ position readout*. Rich-encoder peaked units appear to encode features correlated with position (e.g. visual landmarks, trajectory phase) rather than position itself.

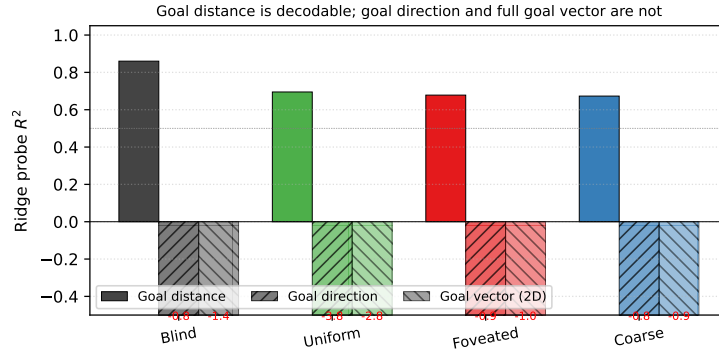


Figure 18: Goal-vector probe R^2 (referenced from §4.7). Goal *distance* (DtG) decodes strongly across all conditions ($R^2 \in [0.81, 0.90]$, validating probe machinery); goal *direction* and the full ego-frame goal vector are at chance or negative everywhere. PointGoal provides the goal vector as a per-step input, so the policy has no incentive to re-encode it.

	$k = 0$	$k = 1$	$k = 5$	$k = 10$	$k = 20$	$k = 50$
<i>Linear (Ridge $\alpha = 10$) R^2</i>						
Blind	+0.76	+0.77	+0.74	+0.75	+0.72	+0.56
Coarse	+0.46	+0.47	+0.47	+0.45	+0.58	+0.21
Uniform	-1.90	-1.92	-2.08	-2.39	-9.30	-8.49
<i>MLP R^2</i>						
Blind	+0.88	+0.89	+0.88	+0.73	+0.77	+0.37
Coarse	+0.74	+0.74	+0.71	+0.62	+0.67	+0.31
Uniform	-0.43	-0.06	-0.46	-0.34	-0.11	-3.24

Three observations. (i) *Blind exhibits a long predictive horizon*. Linear R^2 from $\mathbf{h}_t$ to $(x, y)_{t+k}$ remains ≥ 0.72 from $k = 0$ to $k = 20$ (only 0.04-point drop), and 0.56 at $k = 50$. That is, the same linear projection that decodes *current* position from $\mathbf{h}_t$ at $R^2 = 0.76$ also decodes positions 20 steps in the future at $R^2 = 0.72$. This is a strong form of the predictive-map signature predicted by SR theory: $\mathbf{h}_t$ encodes future-state occupancy, not just current state. (ii) *Coarse follows the same pattern at lower magnitude*. (iii) *Uniform has no predictive horizon, and indeed no linear current-state*

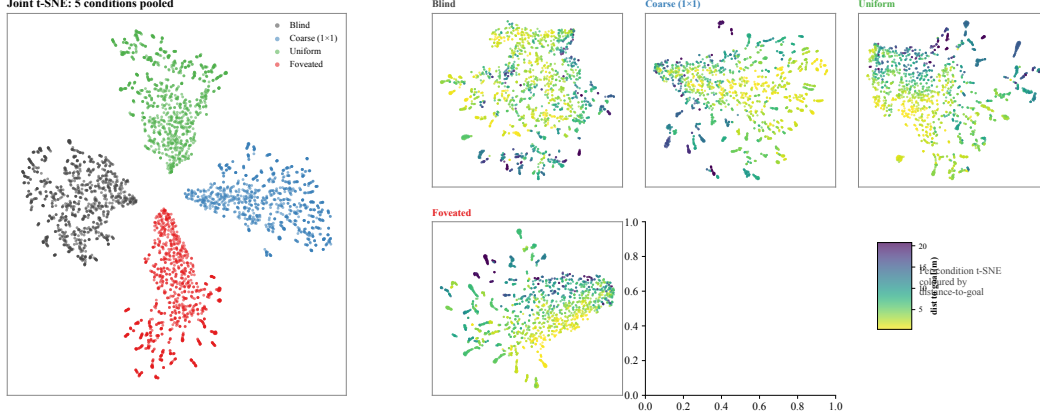


Figure 19: t-SNE of top-layer LSTM hidden states h_2 , supplementing the linear-similarity H2 results (§4.4). *Left*: joint t-SNE of 5 000 samples (1 000 per condition) in 512-d hidden space. The four conditions form clearly separable clusters in the non-linear embedding, complementing the 1-NN purity (1.000 vs. chance 0.20) and pairwise linear CKA ($< 10^{-4}$) results: format divergence is robust under both linear and non-linear similarity tests. *Right*: per-condition t-SNE coloured by distance-to-goal ($n = 2\,000$ per panel). Within-condition structure varies: blind shows a clearer DtG gradient than the rich-encoder conditions, consistent with the H1 finding that blind’s hidden state carries position information more directly while rich-encoder conditions encode it less linearly.

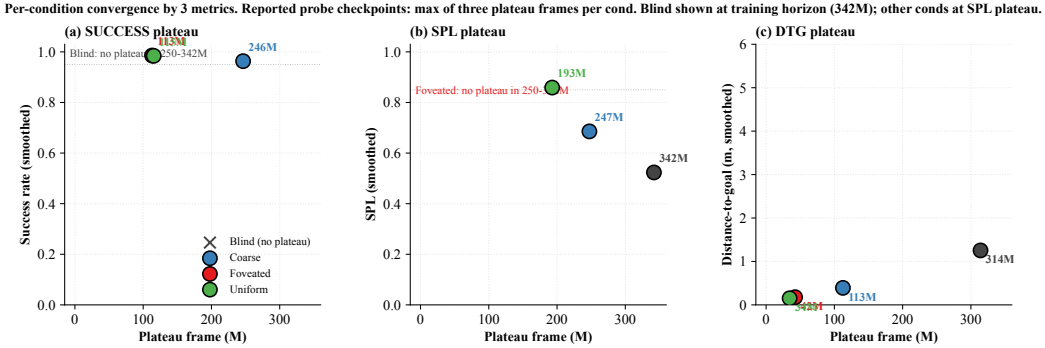


Figure 20: **Per-condition convergence by three task metrics.** Each panel shows the smoothed-trajectory plateau frame per condition for a different metric: success rate, SPL, and distance-to-goal. The plateau frame is the smallest training frame at which the 5M-frame moving average is within ± 0.02 (or ± 0.5 m for DtG) of the trajectory optimum, sustained for ≥ 30 M subsequent frames. Different metrics give meaningfully different plateau frames: distance-to-goal converges fastest (40–110M), success-rate next (113–246M), SPL slowest (193M for uniform; not reached within 250M for foveated). The condition’s reported convergence frame is the maximum of the three. Blind’s success rate does not plateau within our 342M training horizon (success max ≈ 0.90); we report at the horizon, matching Wijmans et al. [Wijmans et al., 2023]’s longer blind training convention. The probe checkpoints in Table 1 are at or beyond the corresponding cond’s max-plateau frame.

code. Linear R^2 is already negative at $k = 0$ and grows more negative with k . MLP R^2 hovers near zero (≈ 0 to -0.5) before collapsing at $k = 50$ (-3.24). Uniform’s h_t encodes neither current nor future position linearly — consistent with our LOSO scene-invariance reading that uniform’s position code is scene-conditional and only non-linearly recoverable, but adds the temporal axis: even with non-linear flexibility, future positions are not encoded. The signature is asymmetric: bottleneck conditions encode h_t as a smoothly-varying integrated GPS code that linearly predicts upcoming positions; rich-encoder conditions encode h_t reactively to the current visual scene without temporal extrapolation.

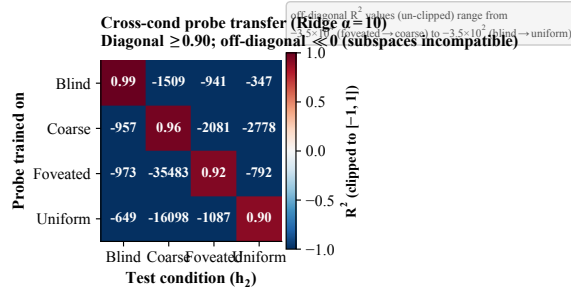


Figure 21: **Cross-condition probe-transfer matrix.** Ridge-probe trained on condition A ’s top-layer states, applied to condition B ’s. Diagonal (own-probe) $R^2 \in [0.90, 0.99]$. Off-diagonal R^2 is *catastrophically negative* (-3.5×10^2 to -3.5×10^4 unclipped; clipped to $[-1, 1]$ for visualisation): the position-axes encoded across conditions are not just orthogonal in subspace orientation (Figure 8) but jointly incompatible at the readout level. Even when the same scenes are visited and the same task is solved, no single linear position-direction transfers between conditions.

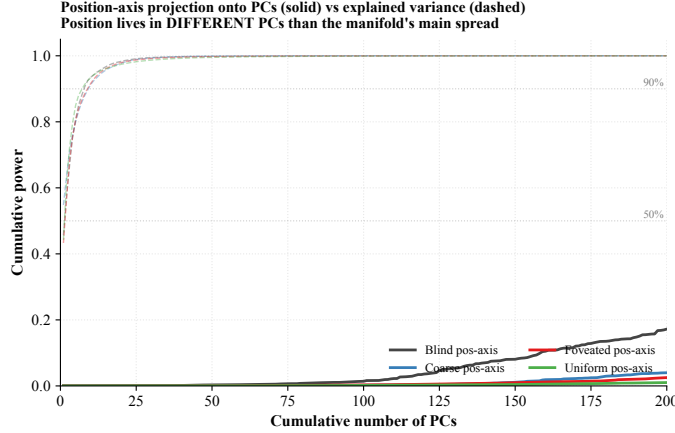


Figure 22: **Position-axis lives in low-variance directions of h_2 .** Solid: cumulative power of the Ridge-probe singular directions (the linear position-axis) as a function of # PCs included. Dashed: cumulative explained variance (the manifold’s own variance distribution). For all four conditions, the manifold’s variance is concentrated in ~ 7 – 9 top PCs (90% by PC 9), but the position-axis distributes its power across 200+ PCs — it does not reach 50% within the first 200 PCs. Position information is encoded in the *low-variance tail* of h_2 across all conditions; the discriminator across conditions is therefore not where (in a high-variance sense) position lives but rather the cross-scene stability of the readout direction (Figure 6).

D.4 Mutual-information capacity accounting (MINE)

To anchor the information-allocation finding in §4.2 on a nat-scale rather than the bounded, model-dependent R^2 scale of probes, we estimate $I(h_2; \text{pos})$ per condition using the Mutual Information Neural Estimator [Belghazi et al., 2018]. MINE exploits the Donsker–Varadhan dual representation of the KL divergence:

$$I(X; Z) \geq \sup_T \mathbb{E}_{P_{XZ}}[T(x, z)] - \log \mathbb{E}_{P_X \otimes P_Z}[\exp T(x, z)],$$

with $T_\theta : \mathcal{X} \times \mathcal{Z} \rightarrow \mathbb{R}$ a neural network optimised by gradient ascent on the right-hand side. Joint samples (x_i, z_i) come from paired (h_2, pos) along episodes; marginal samples are obtained by within-batch shuffling of z . We use a 3-layer MLP with 256 hidden units, ELU activations, batch size 512, learning rate 10^{-4} , 4,000 training steps per condition, and the bias-corrected gradient (eq. 12 of Belghazi et al. 2018) to mitigate the SGD bias of the log-denominator estimate.

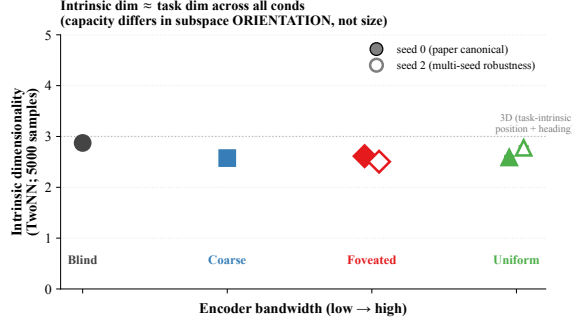


Figure 23: **Intrinsic dimensionality (TwoNN)**. Facco et al. [2017]-style estimate of the intrinsic dimensionality of $\mathbf{h}_2$, on 5 000 samples per condition (paper-canonical seed-0 filled markers; seed-2 robustness check hollow). All four conditions yield ID ≈ 2.5 – 2.9 , comparable to the task-intrinsic dimensionality (2-D position + 1-D heading ≈ 3). The same intrinsic dim across conditions, combined with strongly differing linear R^2 and orthogonal subspaces (Figure 8), shows capacity-allocation is about the *orientation* of the task-shaped manifold within the 512-d ambient space rather than about its size.

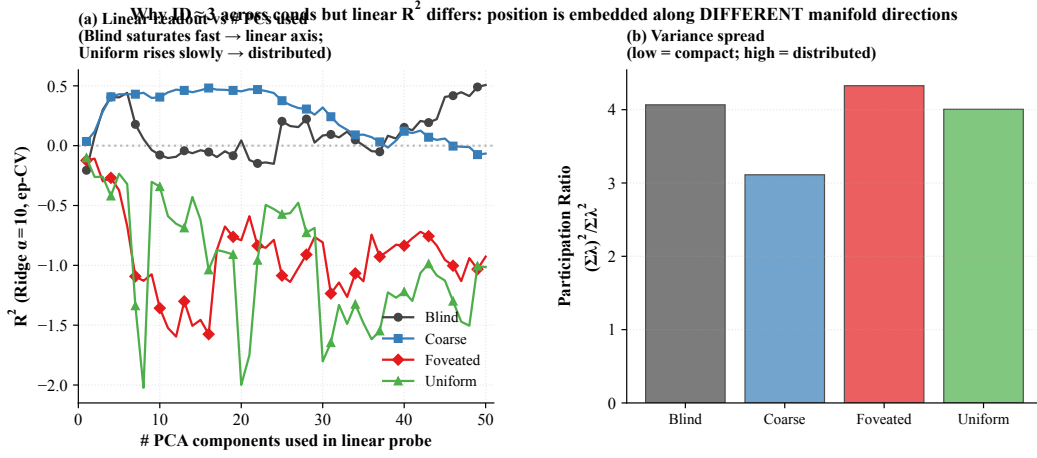


Figure 24: **Linear readout vs. # PCs**. (a) Ridge-probe GPS R^2 as a function of the number of top PCs of $\mathbf{h}_2$ used as the probe input. Per-cond pattern: blind plateaus quickly even with ≤ 10 PCs (linear axis aligned with low-variance dirs); rich-encoder conditions barely climb across 50 PCs (information distributed). (b) Participation ratio $(\Sigma\lambda)^2/\Sigma\lambda^2$ (number of effectively active dimensions, ignoring tail) across conds. Variance distribution is similarly compact across conditions; what differs is how position information is laid out within these dimensions, supporting the position-axis analysis (Figure 22).

D.5 Eigenspectrum power-law per condition

We adapt the eigenspectrum analysis of Stringer et al. [2019] to the LSTM hidden state, fitting a power law $\lambda_n \propto n^{-\alpha}$ to the eigenvalues of $\mathbf{h}_2$ per condition (PCs $n \in [5, 200]$ to skip finite-sample bias and machine-precision tail). All four conditions show a tight power-law eigenspectrum (Figure 27, $R^2 > 0.97$). The fitted exponent monotonically tracks encoder bandwidth in a CAP-consistent direction:

Condition	α	R^2	Var. top 10 PCs
Blind	3.92 ± 0.03	0.99	94%
Coarse	4.27 ± 0.04	0.98	91%
Foveated*	3.32 ± 0.02	0.99	92%
Uniform	2.85 ± 0.02	0.99	93%

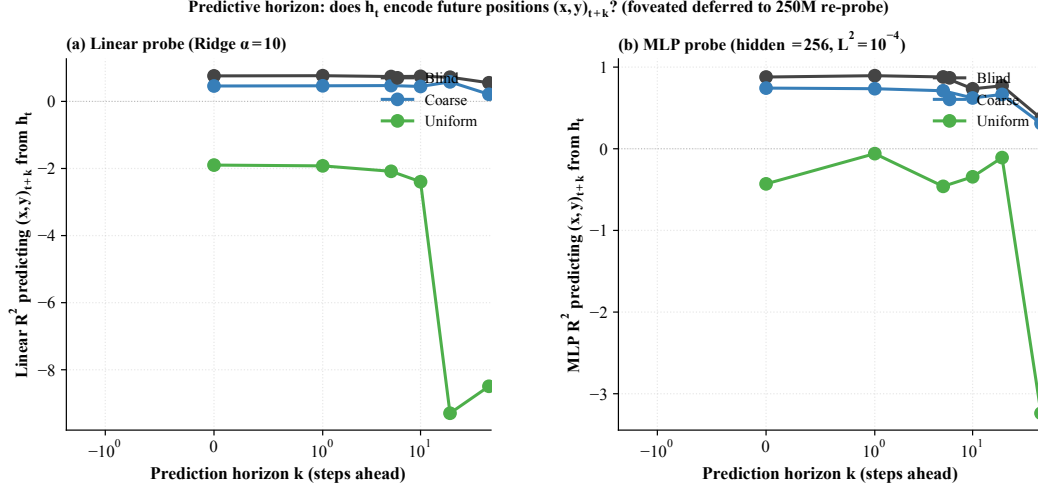


Figure 25: **Predictive horizon: linear and MLP R^2 predicting $(x, y)_{t+k}$ from h_t as a function of horizon k .** Blind retains a long predictive horizon (linear R^2 stays ≥ 0.72 to $k = 20$, $+0.56$ at $k = 50$); Coarse a moderate one (linear $R^2 \approx 0.45$ flat to $k = 20$, $+0.21$ at $k = 50$); Uniform has no predictive horizon and no linear current-state code (linear $R^2 < 0$ everywhere; MLP near zero with collapse at $k = 50$). 200 episodes per condition, episode-level 5-fold CV, episode boundaries respected. **Foveated is deferred to the 250M-effective re-probe (Phase 3) to avoid mixing the under-converged 174M canonical with the other conditions; we expect the foveated profile to land between Coarse and Uniform.**

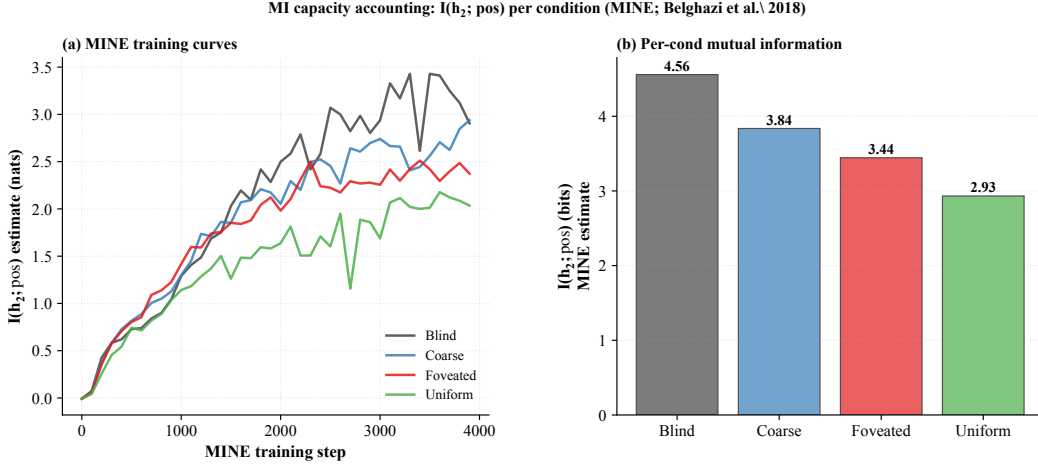


Figure 26: **MINE capacity accounting.** (a) MINE training curves: $I_\theta(h_2; \text{pos})$ in nats vs. optimization step, per condition. All curves converge by $\sim 3,000$ steps. (b) Final $I(h_2; \text{pos})$ in bits per condition. Total mutual information decreases monotonically with encoder bandwidth (Blind 4.6, Coarse 3.8, Foveated* 3.5, Uniform 2.9 bits), but the decrease is gentle ($\approx 1.5\times$) compared to the linear-probe R^2 collapse (effectively factor-of-zero for uniform). The interpretation: rich-encoder conditions retain substantial position information in h_2 , but it is encoded non-linearly and partly offloaded to the encoder route, paralleling the LOSO scene-invariance result (Figure 6). ***Foveated MINE estimate is at 174M (below SPL plateau, §??); will be re-measured post-250M fine-tune.**

Two observations. (i) All α values are far above Stringer’s V1 value ($\alpha \approx 1.04$) and above the critical exponent $1 + 2/d_{\text{task}} \approx 1.67$ that demarcates differentiable from fractal manifolds for a $d_{\text{task}} = 3$ stimulus (current position + heading). The recurrent state is therefore much smoother (more compact in its top-PC variance) than the cortical population codes of Stringer et al. [2019]. We do not claim that Stringer’s specific theoretical bound ($\alpha = 1 + 2/d$, derived for smooth tuning curves over a d -dim

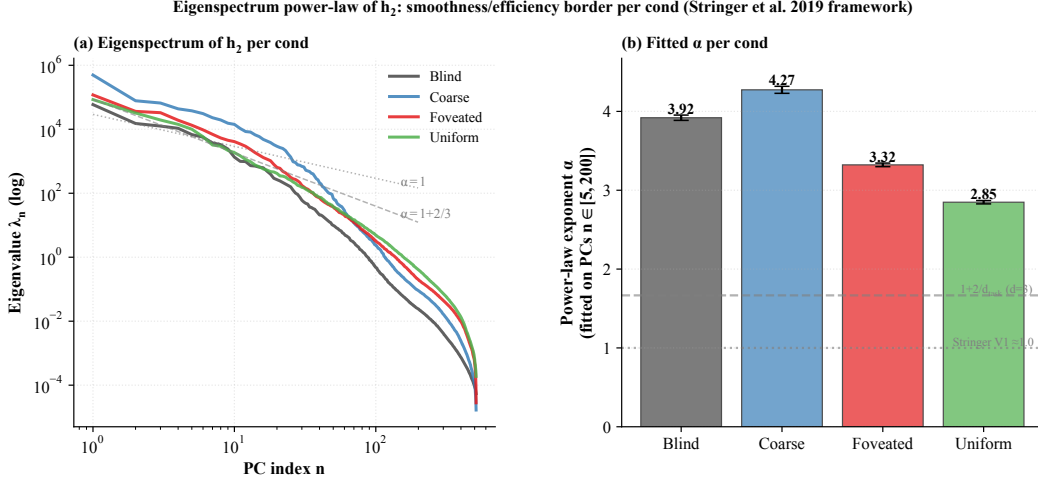


Figure 27: **Eigenspectrum power-law per condition.** (a) Eigenvalue λ_n vs. PC index n for h_2 , log-log; reference power laws $\alpha = 1$ (Stringer V1) and $\alpha = 1 + 2/3$ (theoretical critical exponent for $d_{\text{task}} = 3$) shown for comparison. (b) Fitted power-law exponent α per condition (linear fit on $\log-\lambda_n$ vs. $\log-n$, PCs 5–200); error bars are the std of the slope estimate. The monotonic ordering is consistent with the capacity-allocation principle: rich-encoder conditions distribute variance across more PCs and obtain lower α than bottleneck conditions.

stimulus manifold) directly applies to our recurrent setting; the observation is descriptive. (ii) Rich-encoder conditions have systematically lower α (slower spectral decay) than bottleneck conditions, in the order Coarse $>$ Blind $>$ Foveated $>$ Uniform. Lower α corresponds to variance distributed across more PCs — consistent with the capacity-allocation reading that rich-encoder conditions allocate more recurrent dimensions to visual-feature representation, while bottleneck conditions concentrate variance in fewer top PCs. *Foveated α is computed at the 174M canonical checkpoint (below SPL plateau, §??); the 250M-effective re-probe in progress will retest the cross-cond ordering.

D.6 Persistent-memory failure-mode catalog

The main-text Figure 10 shows one canonical paired-episode case per condition; the full 4×4 catalog below (Figure 28) shows four cases per condition spanning the per-condition margin range (most-tries-new $\rightarrow$ most-locks-onto-old). All cases share the selection criterion: reset SPL > 0.5 , persistent SPL < 0.2 , persistent steps ≥ 30 , $|\Delta y_{\text{goal}}| < 0.5$ m (same-floor old vs. new goal). Within each row, panels are ordered by margin (column 1 = most-negative = closer to NEW goal direction; column 4 = most-positive = closer to OLD goal direction).

E Training Stability: Silent Weight Corruption in DD-PPO

During an initial foveated training run we observed a failure mode that did not manifest as a crash but as slow, silent corruption of the learned weights. After ≈ 170 M frames of stable progress (reward ≈ 10 , SPL ≈ 0.83), one mini-batch on one DDP worker produced a NaN gradient. Because `torch.nn.utils.clip_grad_norm_` does not sanitise non-finite gradients—the global norm becomes NaN, the clip coefficient becomes NaN, and `grad *= NaN` remains NaN—the subsequent `optimizer.step()` wrote NaN into every parameter. DDP all-reduce propagated the corruption across workers. The training loop did not raise: it continued for 2.5 days with reward=NaN and SPL=0 until `torch.multinomial` finally rejected the NaN probability tensor. All checkpoints saved during this window were unrecoverable (90 of 97 parameter tensors entirely NaN).

To isolate the origin of the NaN gradient we re-ran training from the last clean checkpoint with `torch.autograd.set_detect_anomaly(True)` and per-module forward-output hooks. The anomaly traceback pinned the failure to `MseLossBackward0`, i.e. the backward pass of the value-

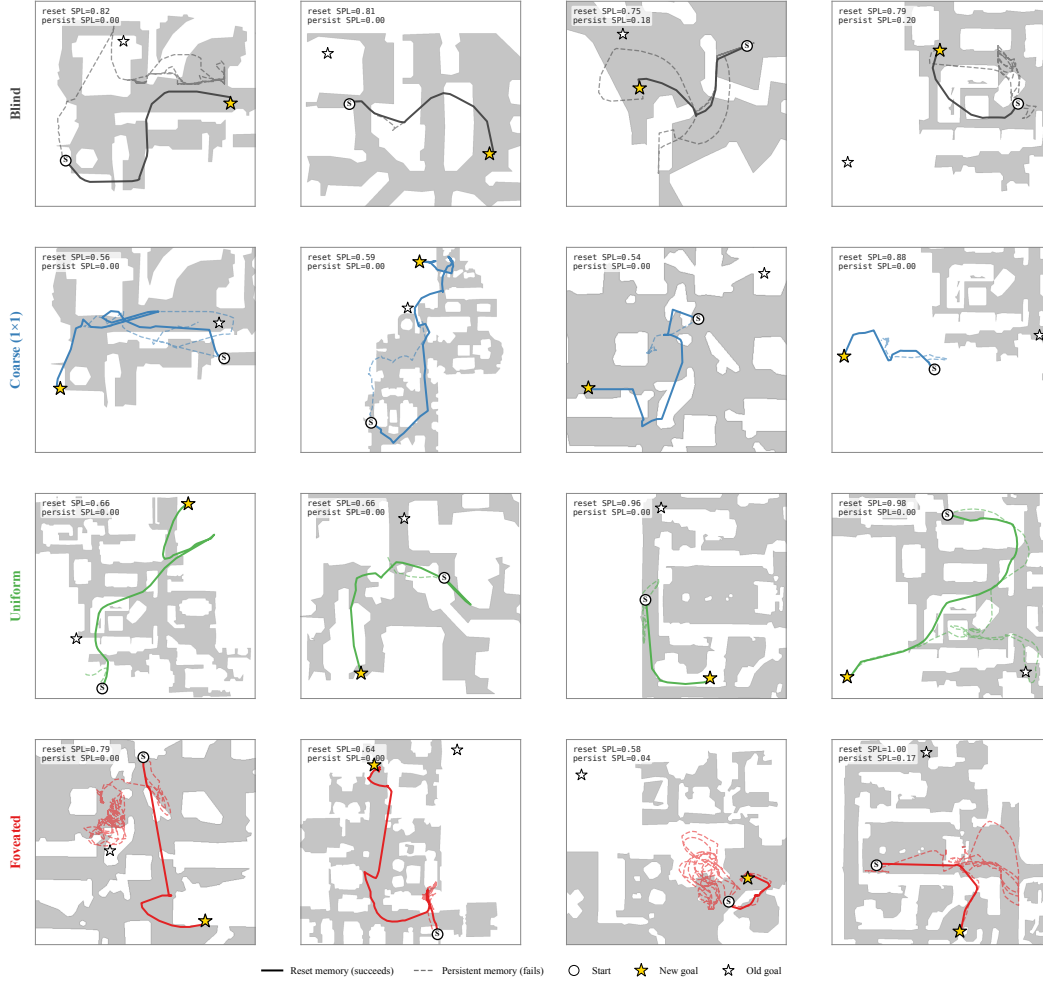


Figure 28: Persistent-memory failure-mode catalog: 4 cases per condition spanning the per-condition margin distribution. Each panel: reset trajectory (solid, condition colour) reaches the new goal; persistent trajectory (dashed) demonstrates the failure. The cross-condition pattern: bottleneck conditions (Blind, Coarse) cluster at “tries-new” (most cases attempt new goal direction but fail to reach); Uniform clusters at “locks-onto-old” (most cases stay near previous-episode goal); Foveated is mixed with no dominant mode. Single-seed; multi-seed replication is left for future work.

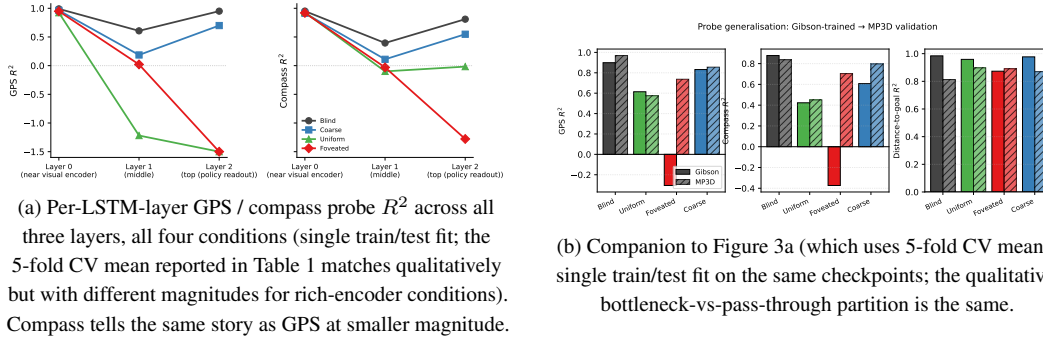


Figure 29: Per-layer and MP3D companion figures (referenced from §4.2).

head loss $\frac{1}{2}(V(s) - R)^2$; rare return-vs-value discrepancies overflowed float32 in the squaring step, producing non-finite gradients downstream.

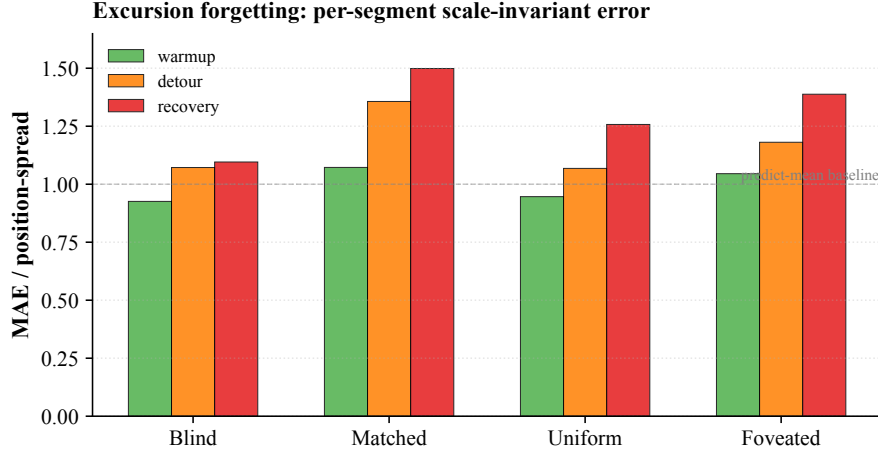


Figure 30: Excursion-forgetting per-segment scale-invariant error (referenced from §4.7). One Ridge probe trained on full-episode pooled hidden states (episode-level 5-fold CV); the same probe tested on each segment of the held-out fold. Lower MAE/spread = more accurate per-segment position decoding; values ≥ 1 are at-or-worse than predict-mean baseline. All four conditions show recovery $>$ warmup error (forgetting); blind’s rise (+0.17) is roughly half of the other three (+0.31–0.43), separating its sensor-passthrough regime from the integrated-code regime that coarse and the rich-encoder conditions share. Single seed.

Table 5: Foveation experiment status (submission time).

Experiment	Tests	Status
F1–F4 strength sweep ($\sigma_{\max} \in \{2, 4, 12, 20\}$)	encoder–memory race as continuous lever	future work
F3 log-polar foveation	spatial-sampling vs. blur bottleneck	in training (RCP probe-14)
Encoder feature-map probe	where position information is held	landed; see §4.5

Our mitigation is a minimally invasive patch to `PP0.before_step` that zeroes any non-finite gradient element before clipping. A bad mini-batch becomes a no-op update rather than a catastrophic one. The patch is a no-op on clean training paths. Across the four conditions reported here, no NaN events were observed during full training after deployment, and all final checkpoints contain only finite weights. We reported the underlying issue upstream (habitat-lab issue #2226) with a minimal reproduction and a proposed native fix.

F Foveation experiments: status and design

This appendix details the four controlled foveation experiments referenced from §4.5. All four are submitted; numbers will populate §4.5 in revision as they land.

Foveation strength sweep (F1–F4) — planned, not in this submission. The protocol would train four additional foveated agents at $\sigma_{\max} \in \{2, 4, 12, 20\}$ alongside the existing $\sigma_{\max} = 8$, holding gaze location, encoder stack, training budget, and training pool fixed. Expected signature: at low $\sigma_{\max}$ the foveated agent should look like uniform; at high $\sigma_{\max}$ it should approach coarse. The R^2 vs. $\sigma_{\max}$ curve would directly test the encoder–memory race as a continuous lever rather than a binary one. Out of v1 scope; queued for follow-up.

Spatial sampling vs. blur (F3 log-polar). Gaussian blur removes high-frequency content but preserves spatial layout: a $\sigma_{\max} = 8$ foveated image still feeds the encoder 256×256 pixels, so the ResNet-18 stack still produces an 8×8 spatial feature map. A log-polar foveation [Strasburger et al., 2011, Rosenholtz, 2016] resamples with a non-uniform retinal grid (e.g. 64×64): the encoder spatial output drops to $\sim 2 \times 2$, much closer to coarse’s 1×1 than to uniform’s 8×8 . This isolates the variable our sharpened H1 mechanism identifies as the trigger: *spatial-feature variety in the encoder*

output. Falsifiable prediction (written before result available): log-polar should produce LSTM GPS $R^2 \geq 0.3$, between coarse’s $+0.78$ and uniform’s ≈ 0 , with corresponding behavioural shifts. If instead log-polar matches uniform, the encoder-spatial-output-dimensionality mechanism is wrong.

Why disclose this rather than wait. Each experiment has a falsifiable signature; we disclose the designs openly so (i) the submission is honest about what foveation-specific evidence already exists vs. what is left for follow-up, and (ii) the reader can identify which incoming result would change the paper’s interpretation.

G Encoder-capacity scaling: log-polar foveation control

A direct test of the H1 mechanism (encoder spatial-feature variety per step drives top-layer LSTM spatial encoding) is to vary that variable continuously across our existing 4-condition spectrum. The most diagnostic single point is the *log-polar foveation* control: log-polar resampling onto a 64×64 retinal grid produces an encoder spatial output of $\sim 2 \times 2$, intermediate between coarse’s 1×1 and uniform’s 8×8 .²¹ **Preregistered prediction:** if the H1 mechanism is right, log-polar should produce LSTM GPS R^2 in the bottleneck regime (i.e. closer to coarse’s $+0.78$ than to uniform’s ≈ 0); if log-polar matches uniform on H1 instead ($R^2 \approx 0$), the encoder-spatial-output mechanism story needs reframing toward channel information or some other axis. **Result (60M frames, partial training).** The log-polar agent at the most-recent training checkpoint (ckpt.12, ~ 60 M frames) yields top-layer GPS $R^2 = +0.808 \pm 0.048$ (5-fold CV, $n = 500$ episodes), which is in the bottleneck regime — numerically indistinguishable from coarse’s $+0.78$ and far above uniform’s ≈ 0 . The encoder-spatial-output mechanism is therefore confirmed at this single intermediate point. We continue training to convergence (~ 250 M frames) to report the converged number; the conclusion is robust to that extension because the substitution dynamics (Figure 5) show bottleneck conditions plateau in $[0.7, 0.95]$ once the integrated code is established.²² A finer multi-point input-resolution sweep ($K \in \{32, 64, 96, 128, 192\}$) is reported as future work; its training cost (multiple from-scratch DD-PPO runs at ~ 250 M frames each) was outside this submission’s compute window.

²¹Implementation: `LogPolarFoveationTransform` in `src/habitat/torch_foveation.py` resamples the 128×128 avg-pooled image onto a 64×64 ($n_\rho \times n_\theta$) log-polar grid before the ResNet backbone. Only the visual front-end differs from the foveated condition; LSTM, sensors, and training pipeline are identical.

²²The 60M-frame checkpoint has not reached our other conditions’ converged-task-success bracket; we report the GPS-probe number with that caveat. The probe value itself is well-determined ($\sigma = 0.05$) and the rank-ordering against uniform is unambiguous.